Lean Boot Camp

The Five P's of a Lean Transformation

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Other Modules In This Series:

5S

Standard Work

Total Productive Maintenance (TPM)

Administrative Lean

Cell Design/ Production Preparation Process (3P)

Kaizen

Kanban

Problem Solving

Poka Yoke Quality Systems

Transaction Process Improvement (TPI)

Cause & Effect Diagram with the Additions of Cards (CEDAC)

Single Minute Exchange of Dies (SMED)

Value Stream Mapping (VSMi®)

Visual Management

Managing Daily Process Excellence



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Strategically Creating Value Through the Elimination of Waste

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Ground Rules for Event

- Safety first (use personal protective equipment)
- Active participation by all
- Start and end on time
- All electronic gadgets (phones, pagers, etc) turned to “off” or “silent”
- Computers to be used only when instructed
- Keep an open mind to change
- Don’t let “perfect” get in the way of “better”

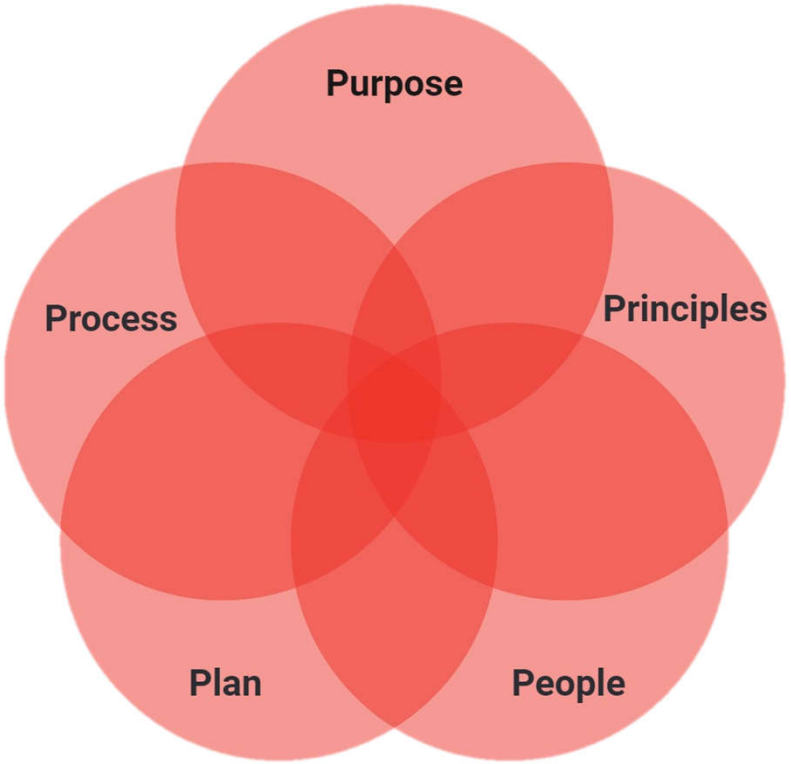
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Objectives

- Introduce and/or reinforce the framework of a Lean transformation.
- Compare and contrast your organization through a Lean lens.
- To stimulate dialogue to discover the compatibility of the Lean framework with your organization.
- Understand your role as a leader in a Lean transformation.
- Develop non-negotiables as they relate to the Lean framework.

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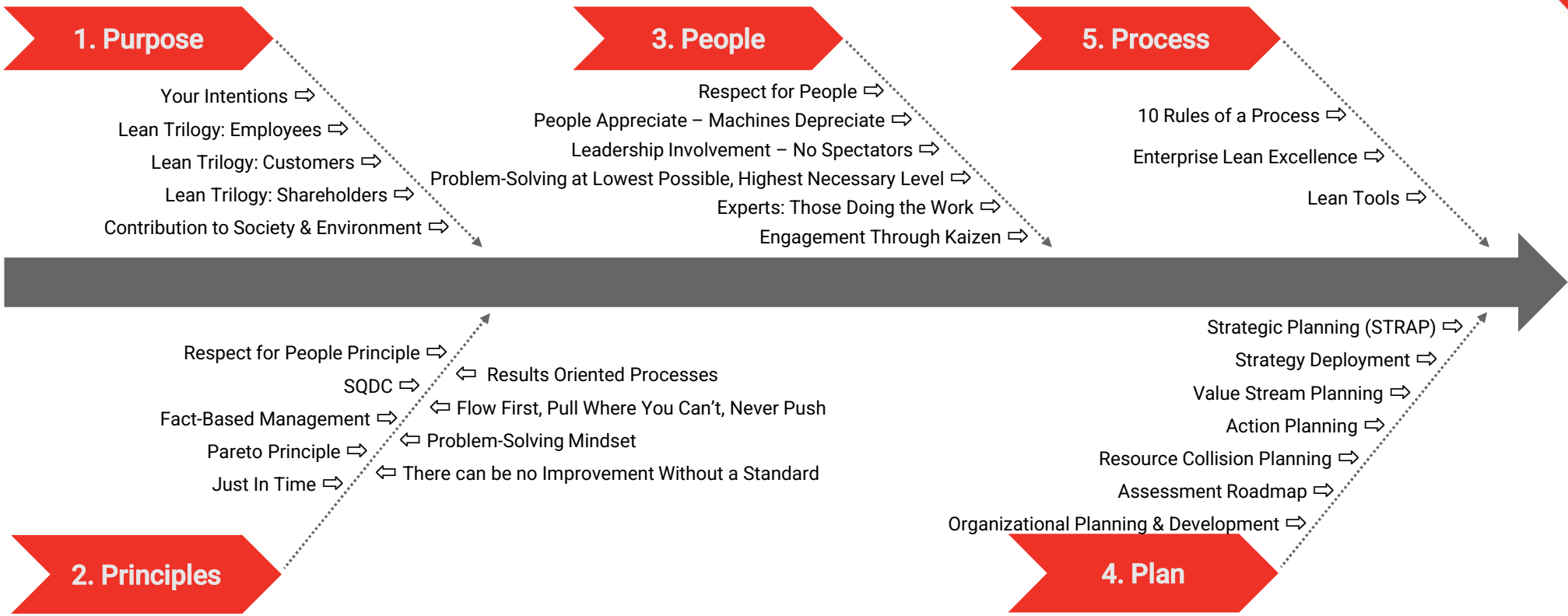
The Five P's and Profitable Growth



= Profitable Growth

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Lean Boot Camp Framework

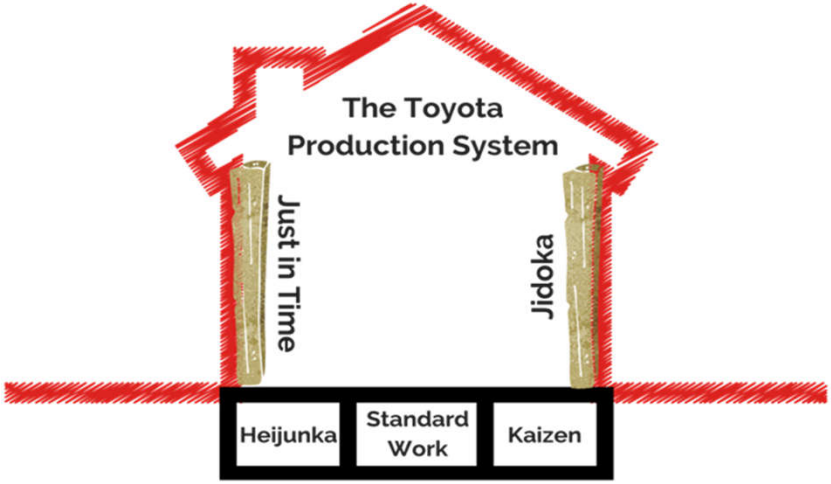


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TPS House – Lean Basics



Father of the Toyota Production System
Taiichi Ohno
1912 - 1990



"If you are going to do TPS, you must do it all the way…"
- Taiichi Ohno



Head Coach of the Green Bay Packers
Vince Lombardi
1913 - 1970

"Gentlemen, this is a football."
– Vince Lombardi

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Purpose

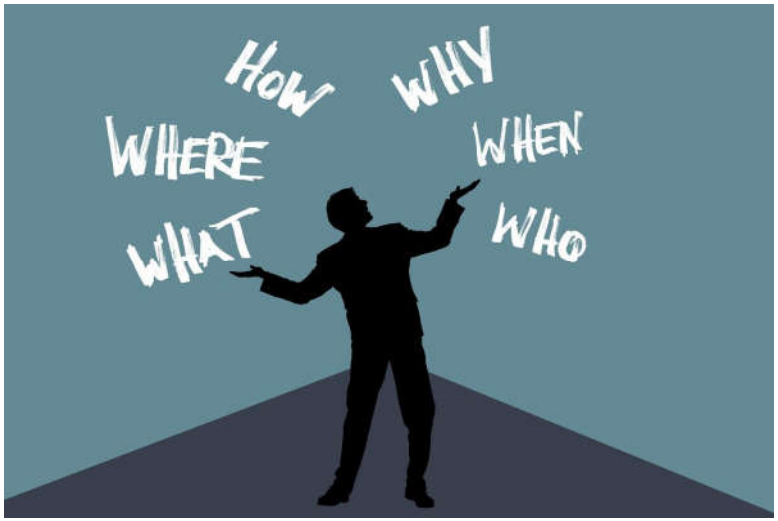
- Your Intentions
- Lean Trilogy:
  - Employees
  - Customers
  - Shareholders
- Contribution to Society & Environment

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Your Intentions

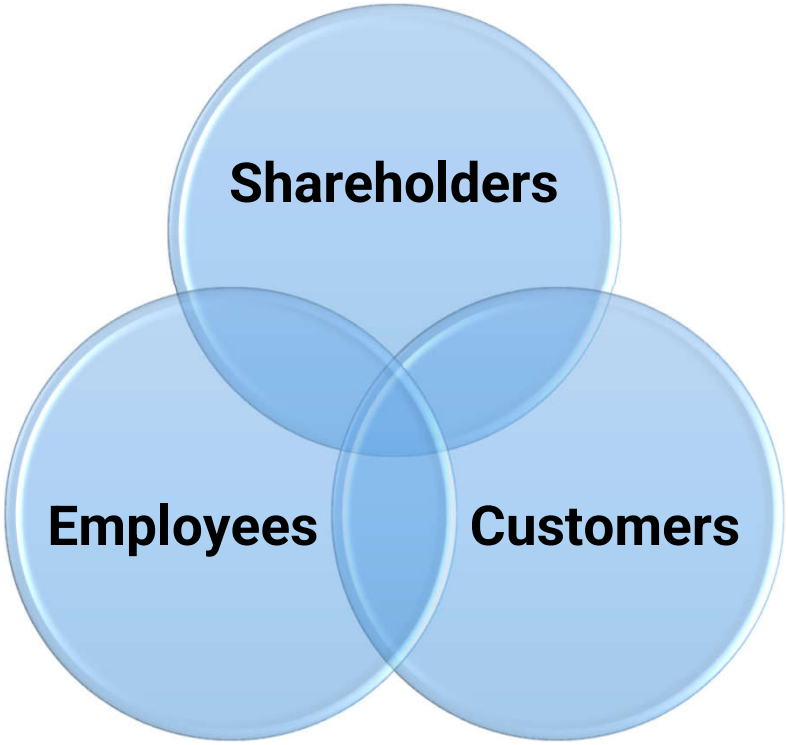
Discussion:

1. Why do you want to embark on a Lean transformation?
2. Who would care about your reasons for doing so?



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Lean Trilogy



All stakeholders must win.

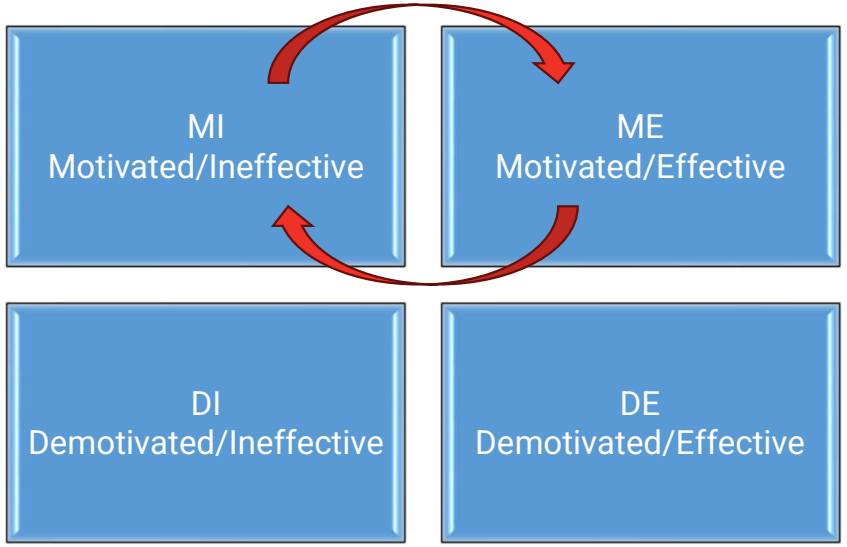
A Lean Transformation cannot be a zero-sum game.

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Lean Trilogy: Employees

- Empowerment and Engagement:
  - Lean is about people. Engage employees with the WIFM (What's In It For Me?) factor.
  - Develop processes that enable ordinary employees to achieve extraordinary results.
- Continuous Improvement:
  - Encourage participation in Kaizen events.
  - Focus on process improvement, not individual faults.
- Motivation and Morale:
  - Recognize and reward positive changes.
  - Provide training and support for all employees.

Where is your organization today?



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Lean Trilogy: Employees

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. What will you communicate to your employees as to the reasons why you are embarking on a Lean transformation?
  2. Why should your employees care?



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Lean Trilogy: Customers

- Voice of the Customer (VOC):
  - Have you asked your customers what is important to them?
  - Don’t assume—engage and listen to customer needs.
- Responsiveness:
  - Address customer inquiries and requests promptly.
  - Align services and products to exceed customer expectations.
- Customer Value:
  - Customers may perceive value on criteria other than your product or service.

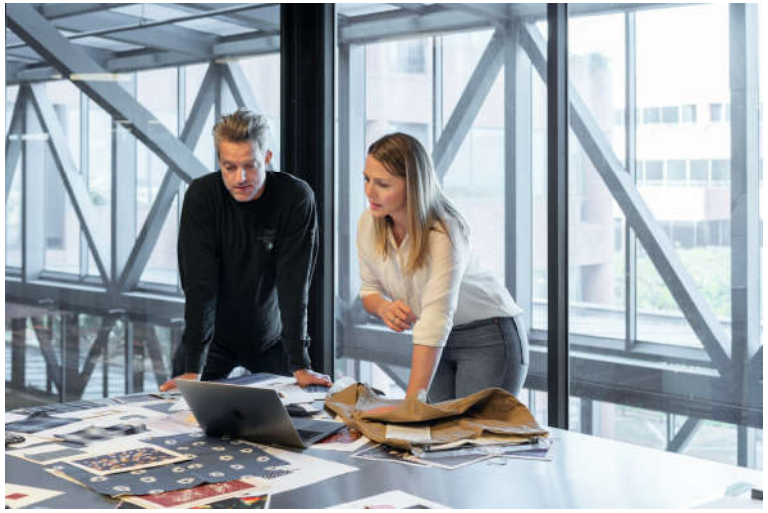


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Lean Trilogy: Customers

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. Who is/are your customer(s)?
  2. Do you know your customer(s) expectations and requirements of you?
     a) How do you know you know?



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Lean Trilogy: Customers

Case Study in Voice of the Customer:

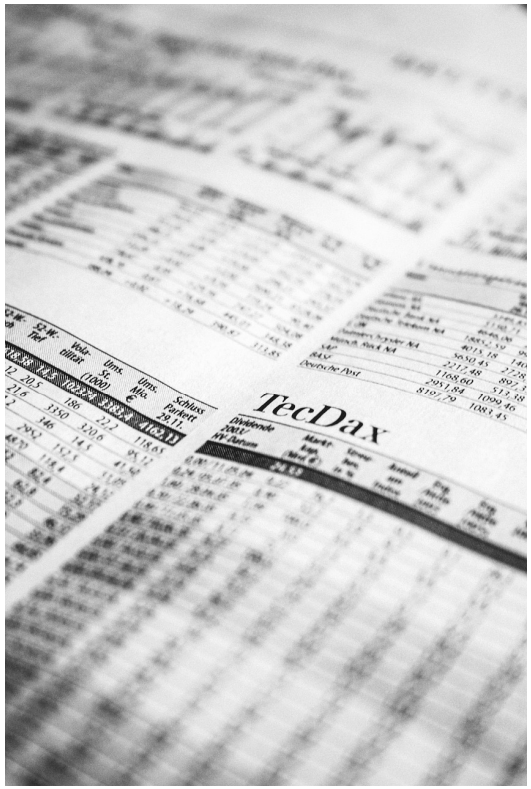
- Jake Brake was rated #106 out of 110 suppliers by Hino Motors.
- Despite 100% on-time delivery & quality.
- How can this be?



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Lean Trilogy: Shareholders

- Balanced Approach:
  - Meeting employee and customer needs leads to shareholder satisfaction.
  - Optimize the balance between the three stakeholder groups.
- Long-Term Success:
  - Focus on sustainable growth through operational excellence.
  - Avoid short-term cost-cutting measures that harm employee and customer relationships.
- Trade-Off Management:
  - Make informed decisions considering all stakeholder impacts.
  - Example: Pricing strategies that balance customer satisfaction and profitability.



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Lean Trilogy: Shareholders

Honda Prioritizes:
Employees 1st
Customers 2nd
Shareholders 3rd

If we do not take care of our **employees**

They will not take care of our **customers**

If we do not take care of our **customers**

We will not take care of our **shareholders**



Case Study Example:
Honda Marysville, OH

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Lean Trilogy: Shareholders

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. What does the shareholder value?
     a. How will you know?
  2. How does this interact with customers and employees?



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Contribution to Society & Environment

- Job Creation and Economic Growth:
  - Increased profits enable hiring more employees, boosting local economies.
  - Lean efficiency can lead to cost reductions, making products more affordable for consumers.
- Environmental Stewardship:
  - Reducing waste minimizes the amount of scrapped material ending up in landfills.
  - Less waste requires fewer raw materials, which means less of that material is being removed from the environment (wood, metals, etc.).
- Operational Excellence:
  - Reduced defects and rework will lead to less waste and better product quality.
  - Better products lead to better product reliability and customer trust through consistent quality.



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Purpose | Principles | People | Plan | Process

Purpose - Recap

- Your Intentions
- Lean Trilogy:
  - Employees
  - Customers
  - Shareholders
- Contribution to Society & Environment

What were your key takeaways from this section?

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Principles

- Respect for People Principle
- SQDC
- PARETO Principle
- Just In Time
- Results Oriented Processes
- Flow First, Pull Where You Can't, Never Push
- Problem Fixing vs. Problem-Solving
- Daily Management vs. Strategic Deployment
- The can be no improvement without a STANDARD
- Fact-based Management

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Principles

Principles and values without an underlying process are just slogans.

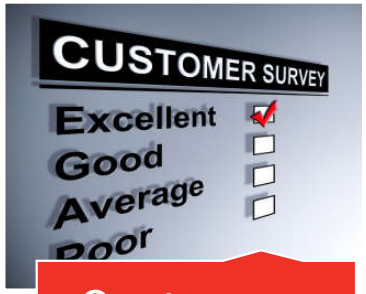


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Respect for People Principle



Employees



Customers



Shareholders



Community



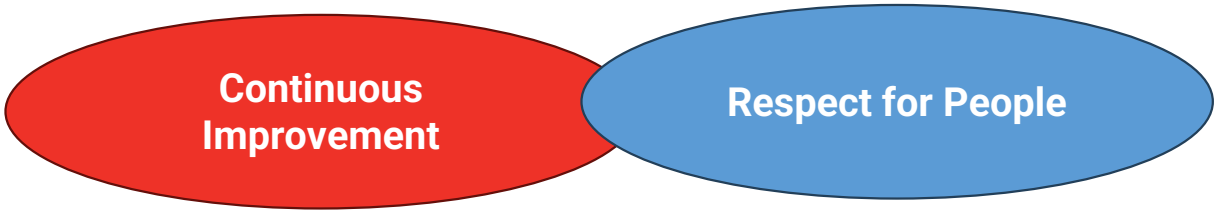
Competition



Environment

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Continuous Improvement and Respect for People



Challenge

We form a long-term vision, meeting challenges with courage and creativity to realize our dreams

Kaizen

We improve our business operations continuously, always driving for innovation and evolution

Genchi Genbutsu

Go to the source to find facts, make correct decisions, build consensus & achieve goals at the best possible speed

Respect

We respect others, make every effort to understand each other, take responsibility and do our best to build mutual trust

Teamwork

We stimulate personal and professional growth, share the opportunities of development and maximize individual and team performance

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Respect for People Principle

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. What does Respect for People mean to you?
  2. How well do you think your company practices the RFP Principle?



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Safety, Quality, Delivery, Cost (SQDC) Hierarchy

- Should be consulted for all decisions.
- An order of priority, not a list.
- Metrics should reflect SQDC.
- Negative consequences will result from violating the SQDC hierarchy.



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Safety, Quality, Delivery, Cost (SQDC) Hierarchy



	Item	Expectation
1	**Safety**	***MINIMUM*** 50% improvement per year.
2	**Quality**	***MINIMUM*** 50% improvement per year.
3	**Delivery**	Lead-time 50% better than competition. Strive for 100% On-Time Delivery to Customer ***Request*** Date with a minimum of 50% annual lead time reductions.
4	**Cost**	Productivity improvements 2% per month for the first 24 months, 1% per month thereafter. Double-digit inventory turns each year for the first 3 years. CAPEX <2% of revenue.
=	**EPS Growth**	10-15% core EPS growth per year (without acquisitions)

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SQDC – Would you…?

Safety

- **Would you ship an unsafe product to meet an urgent customer delivery date?**

Quality

- **Would you delay shipment until a suspect quality issue is fixed, even if it would shut down a customer's production line, or ship it as-is?**

Delivery

- **Would you ship via ground shipping to maintain profitability or ship air freight to meet on-time delivery commitments?**

Cost

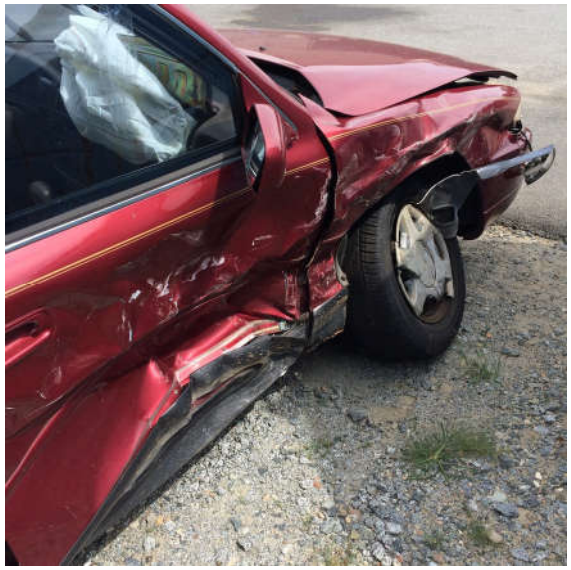
- **How do you think safety, quality, and delivery affect cost?**

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Fact-Based Management

- **Fact**: 100% of all auto accidents involved cars with black tires.
- **Conclusion**: Do not drive cars with black tires.

Do you agree or disagree?



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Fact-Based Management

"In God we trust. All others bring data."

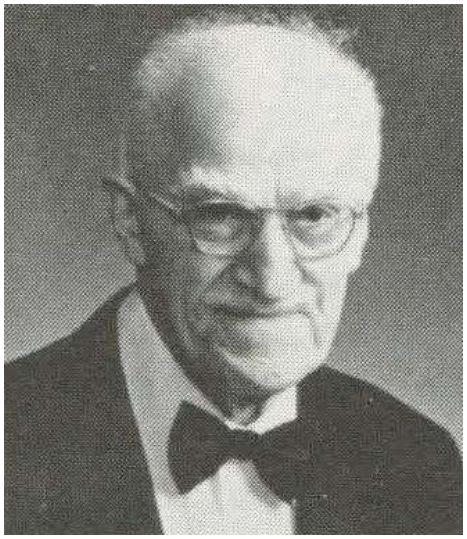
– *W. Edwards Deming*



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Pareto Principle

The Pareto Principle, also known as the 80/20 rule, suggests that roughly 80% of effects are caused by 20% of the causes.

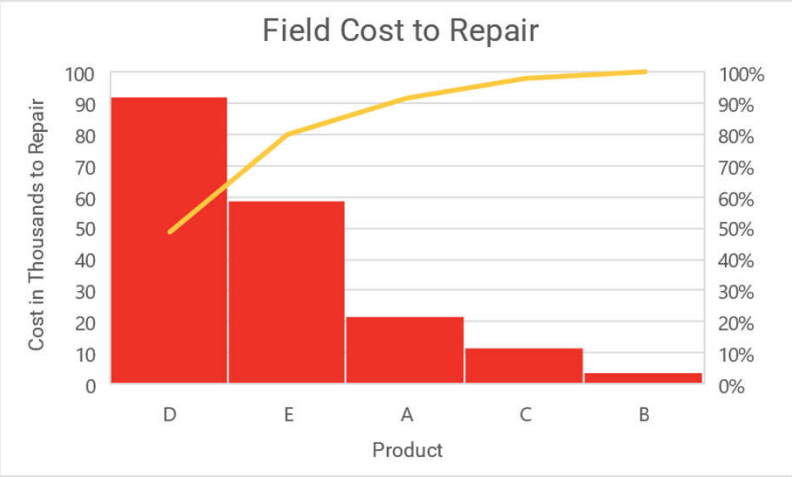


Dr. Joseph Juran

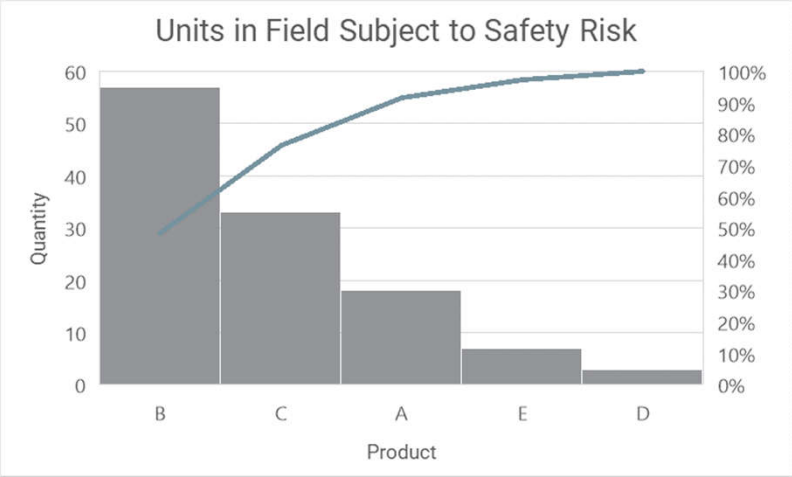
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Pareto Principle

Cost to Repair Quality Issues



Safety Issues

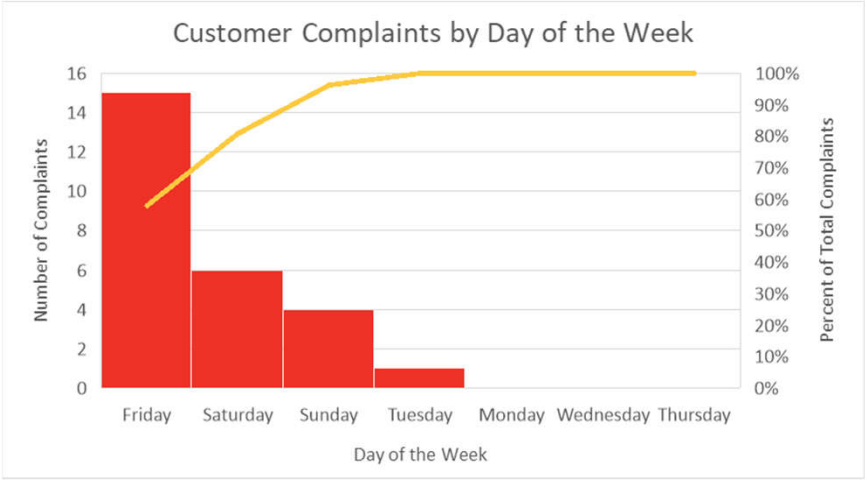


Which product's problems would you attack first?

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Pareto Principle – Restaurant Example

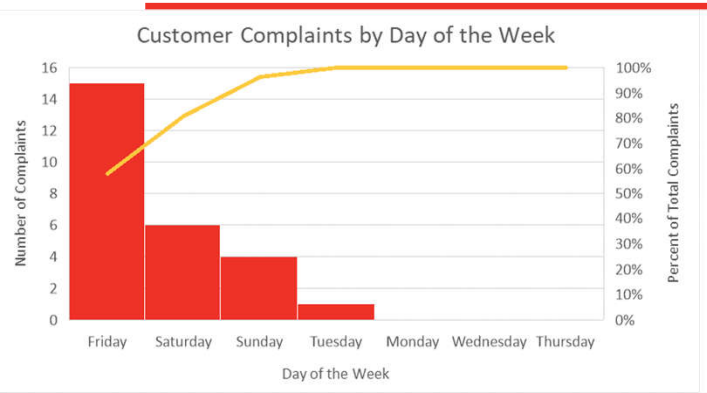
A restaurant owner recognized they were getting more customer complaints on certain days of the week than others.

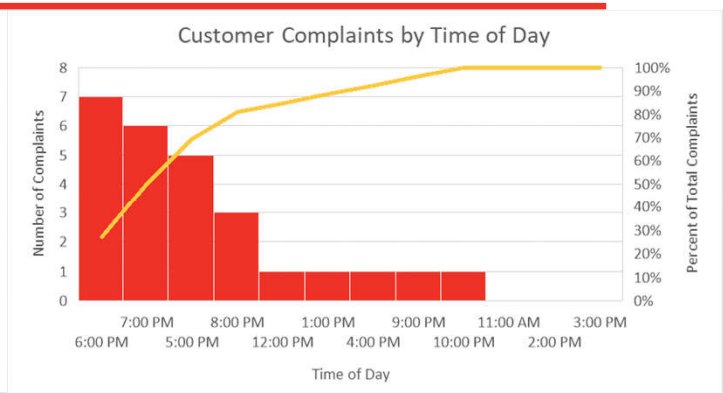


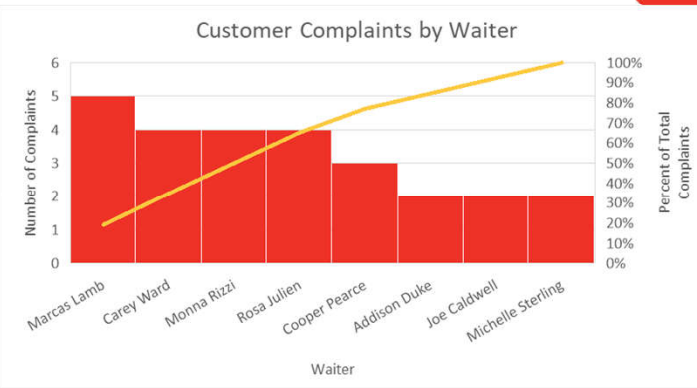
What other types of data would you Pareto to better understand customer complaints?

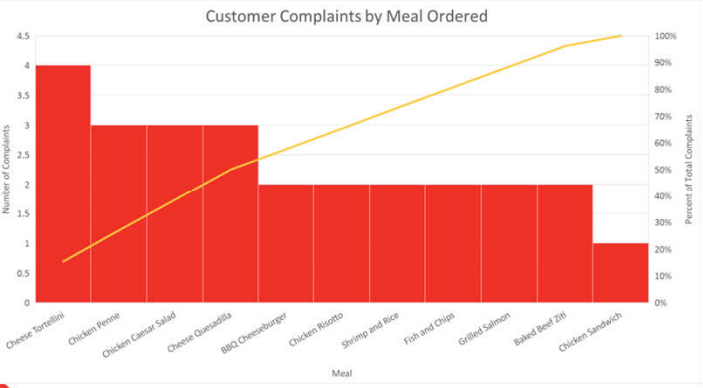
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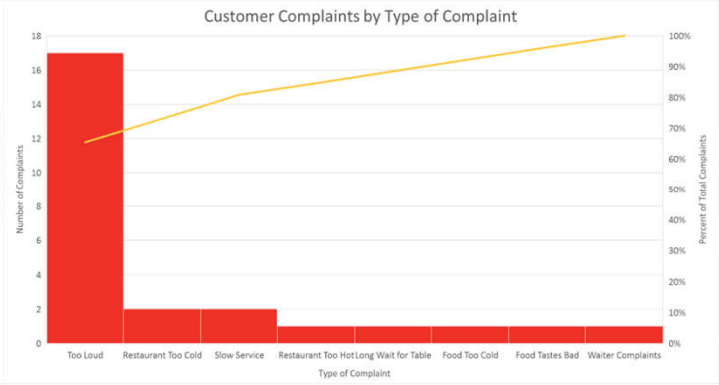
Pareto Principle – Restaurant Example

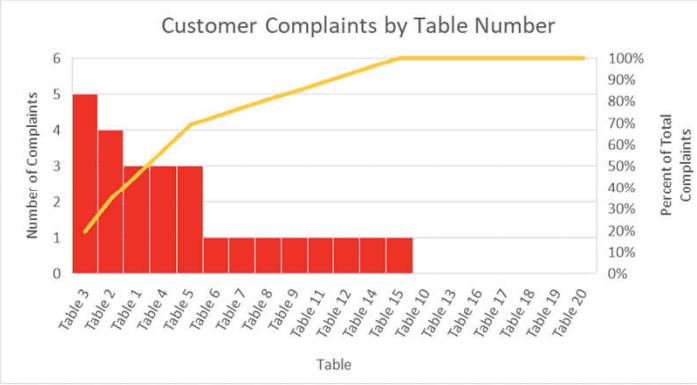












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Just In Time (JIT)

- Just In Time (JIT) focuses on producing and delivering products and services:
  - (S) Safely
  - (Q) With the required quality
  - (D) Exactly when needed and the right quantity
  - (C) At the right cost
- Benefits of JIT:
  - Improved Quality
  - Improved Delivery and Lead Time Performance
  - Enhanced Efficiency and Flexibility
  - Reduced Inventory Costs



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Just In Time (JIT)

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. Does the JIT principle apply to your business?
  2. Does your company operate based on JIT principles?
     a. If no, why not?
  3. Is lead time a competitive advantage in your market?



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Just In Time (JIT)

"The more inventory a company has, the less likely they will have what they need."

– *Taiichi Ohno*



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Results-Oriented Processes

- **Management by Objectives (MBO)** is a process in which a manager and an employee agree on specific performance goals and develop a plan to reach them.
- **Results-Oriented Processes** are high-performing processes designed to deliver repeatable and sustainable business results.

What do you think is the difference between the two?

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Results-Oriented Processes

"Tell me how you measure me, and I'll tell you how I will behave."

— *Eliyahu M. Goldratt, Critical Chain*



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Results-Oriented Processes vs. MBO



K = Kaizen
S = Sustain Results With Standardized Processes

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Results-Oriented Processes vs. MBO

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. Does your company reward results regardless of the process that produces those results?
  2. Are those results sustainable?
  3. For future consideration:
     a. Does the rewards system need to be modified to align with the Lean framework?



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Decoding the DNA of the Toyota Production System

Breakout Discussion:

- Compare and contrast “Decoding the DNA of the Toyota Production System” with your organization.


Decoding the DNA of the Toyota Production System

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Flow First, Pull Where You Can't, Never Push



Rule #1
Flow First

Rule #2
Pull Where You Cannot Flow





Rule #3
Never Push!

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One-Piece Flow

"Kanban is an admission of failure to do one-piece flow."

- Chihiro Nakao





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Flow Exercise – Pipe Cleaner

Instructions:

- Break the larger group into teams.
- Each team gets the same supplies.
- Follow the facilitator's instructions to complete the exercise.



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Problem-Solving Mindset

Problem Fixing

- Provides a temporary correction.
- May not always be able to address the root cause.
- Issue is likely to recur.

Problem-Solving

- Permanently remedies the problem.
- Identifies and eliminates the root cause.
- Prevents recurrence of the issue.

What makes a world-class company is their ability to solve problems. Even the great Toyota has problems…but we only solve our problems once.
- Toyota Executive April 1990

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Problem-Solving Mindset

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. How often do problems reoccur in your business?
     a. Could you provide examples?
  2. Are we problem fixers or problem solvers?
  3. How much time would we make available if we could problem-solve rather than problem-fix?



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Standardization

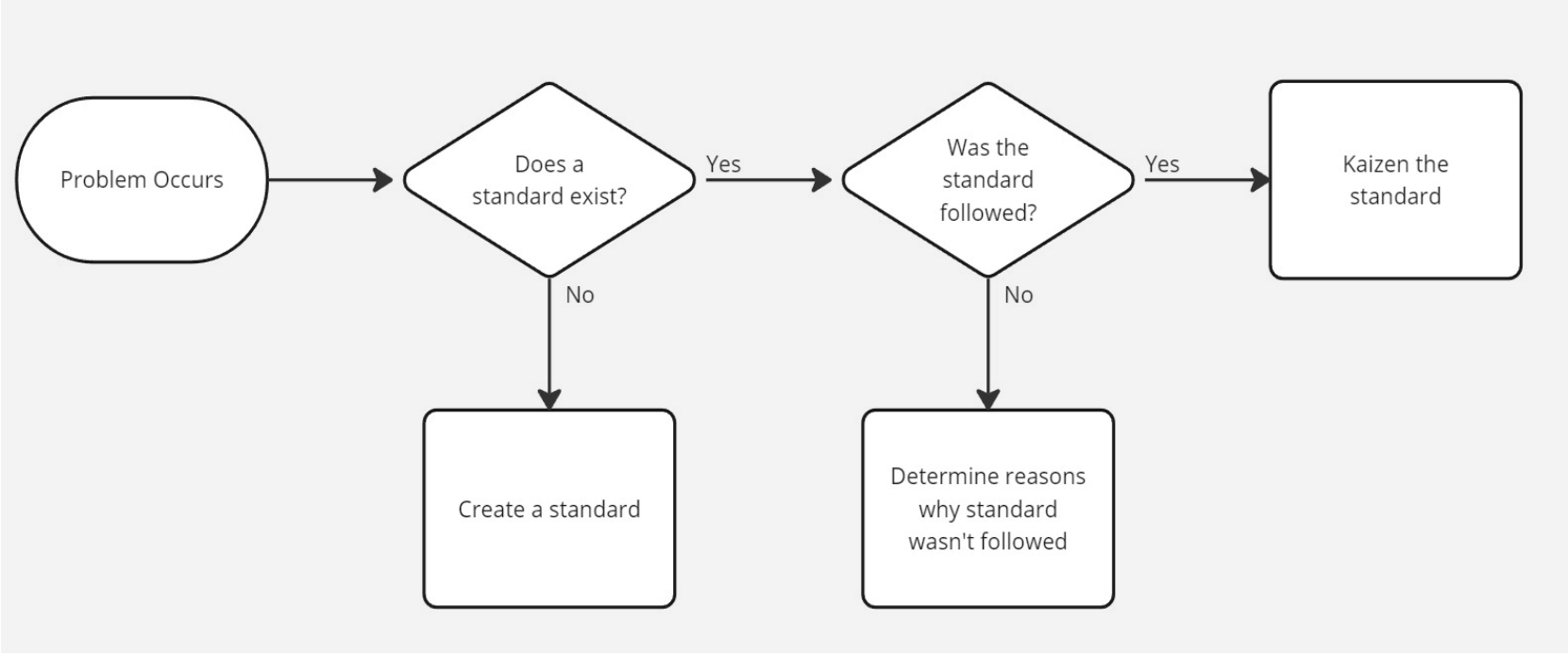
"There can be no Improvement Without a Standard."

– *Taiichi Ohno*



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There Can Be No Improvement Without a Standard



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Standardization

"If you cannot describe what you're doing as a process, then you don't know what you're doing."

– W. Edwards Deming



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Numbers Exercise

Instructions:

- Break the larger group into teams.
- Each team gets the same supplies.
- Follow the facilitator's instructions to complete the exercise.



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Principles

- Respect for People Principle
- SQDC
- PARETO Principle
- Just In Time
- Results Oriented Processes
- Flow First, Pull Where You Can't, Never Push
- Problem Fixing vs. Problem-Solving
- Daily Management vs. Strategic Deployment
- The can be no improvement without a STANDARD
- Fact-based Management

What were your key takeaways from this section?

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People

- Respect for People
- People Appreciate – Machines Depreciate
- Leadership Involvement – No Spectators!
- Problem-Solving at Lowest Possible, Highest Necessary Level
- Experts: Those Doing the Work
- Engagement Through Kaizen

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Respect for People

- **Respect for People Principle:** Rooted in the Toyota Way, emphasizing the importance of respecting and valuing all stakeholders.
- **Employee Engagement:** Kaizen promotes active involvement and ownership.
- **No-Layoff Policy:** Ensures job security, enhancing morale and trust within the organization.
- **Human-Centered Approach:** Focuses on developing people through training and empowerment.
- **Blameless Environment:** Focus on fixing broken processes rather than blaming individuals. Most problems are a result of bad processes.



Did this change your thoughts from our previous discussion on Respect for People?

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People Appreciate, Machines Depreciate

People

- Gain experience and skills.
- Become more motivated.
- Increase effectiveness and productivity.
- Foster innovation and creativity.
- Contribute to continuous improvement.
- Develop problem-solving abilities.

Machines

- Experience wear and tear.
- Require regular maintenance.
- Depreciate in value over time.
- Have limited adaptability.
- Can break down unexpectedly.
- Need eventual replacement.



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Leadership is not a Spectator Sport

- Active participation on Kaizen teams.
  - Shows commitment to the Lean transformation.
  - The only way to truly learn.
- Regular visits to Gemba.
  - Go See
  - Ask Why
  - Show Respect



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Leader Standard Work Example

Item	Description	Timing (● = Preferred, ❍ = Optional)				
		Daily	Weekly	Monthly	Quarterly	Annual
Kaizen Reviews	Review of strategic Kaizen initiatives with project teams.			●		
SD Matrix Review	Review 1st and 2nd Levels of Business Unit SD matrices, playing "catchball" with Strategic Breakthrough Objectives, AIPs, Targets, etc.				●	
SD Review Sessions	Review of Business Unit progress on SD initiatives. To include reviews of Bowling Chart metrics, Countermeasures and Action Plans.			●		
Operating Review	Conduct Operating Review (different than SD Review) pertaining to the following: Monthly Financial Results, Talent Development, M&A Status, Other Key Operating Issues.			●		
Gemba Walks	Random walks around the "Gemba" to review progress on KPI's. (Note: Daily Management must be deployed first.)		●			
CEO Town Hall	20 Minute discussion with identified High Potential individuals within Arch.		●	❍		
Kaizen Participation	Fulltime participation on selected Kaizen team.				●	
Kaizen Reviews	Participate in Kaizen Report-outs.		●	❍		
Teach/Facilitate Kaizen Event	Teaching/Facilitating Kaizens demonstrates a commitment to Lean, Leadership, and an unbelievable opportunity to learn. Participation must have been first experienced in a hands-on Kaizen event.				❍	●
Status Review	Progress review of status with the Chief Transformation Officer.		●			
Learning Sessions	Participate in Learning sessions with members of the Team. (Format: Pre-read on a selected topic and have discussion sessions in order to expand Leadership learnings.)			●		
Boot Camp	Boot Camp to be attended by the CEO and Team in order to facilitate continued learning and development.					●
Steering Committee	Attend Steering Committee with members of the team. Chaired by the CTO, status report, issues, and action items to be discussed. Agenda and process TBD.			●		
Values Audit	Discussion with the Team of the successes and challenges relating to how the business performed in accordance with their stated values.			●		

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Problem-Solving at Lowest Level Possible, Highest Level Necessary

- Encourages employees to take ownership.
- Quality of the solution is greatly enhanced.
- Fosters a culture of proactive problem-solving.
- Speeds up decision-making and reduces dependencies on management.



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Problem-Solving at Lowest Level Possible, Highest Level Necessary

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. Do you provide employees time, resources, and training to problem-solve?



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Experts: Those Doing the Work

How do you engage your people in your company, and where can that process be improved?



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Engagement Through Kaizen

Attitude of employees towards:	Number of Lean Events:			
	0	1-2	3-5	6+
Supervision	-3	3	8	13
Employee Involvement	-4	3	14	16
Career Opportunities	-1	0	7	13
Job Satisfaction	-2	0	11	14
Customer Focus	-3	2	12	14
Kaizen	-6	7	14	27

Vermeer Manufacturing Employee Survey summary. Note: Data as presented at Shingo Prize Conference 2004

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Employee Engagement

LEAN ≠ Less Employees Are Needed

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. Are you willing to guarantee no layoffs due to freed-up headcount resulting from Kaizen improvements?
  2. What will you do with the freed-up resources that result from your Lean kaizen efforts?



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Deployment Options for Freed-Up Resources

Possible Options:

- ❑ Redeploy to Kaizen teams.
- ❑ Redeploy to other areas of the shop.
- ❑ Train freed-up resources to gain multi-skills.
- ❑ Use freed-up resources to cover for absenteeism and vacations.
- ❑ Insource operations that strategically make sense to your business and/or fit your core competency.
- ❑ Create a flex schedule, which optimizes your overall workforce.
- ❑ Let attrition take its course.
- ❑ CHALLENGE YOUR SALES FORCE TO FILL THIS EXCESS CAPACITY.



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People

- Respect for People
- People Appreciate – Machines Depreciate
- Leadership Involvement – No Spectators!
- Problem-Solving at Lowest Possible, Highest Necessary Level
- Experts: Those Doing the Work
- Engagement Through Kaizen

What were your key takeaways from this section?

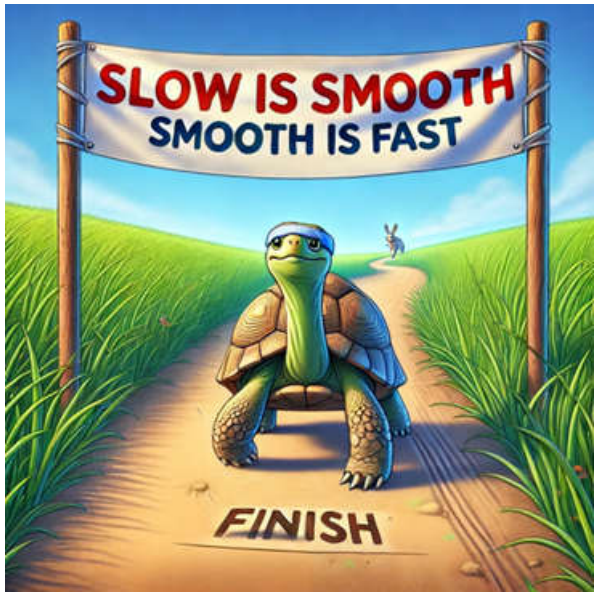
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Plan

- Strategic Planning (STRAP)
- Strategy Deployment
- Value Stream Planning
- Action Planning
- Resource Collision Planning
- Assessment Roadmap
- Organizational Planning & Development

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Plan



What does this mean to you?

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The Value of Planning

Toyota's New Product Development Cycle = 18 months



Results in improved quality and less rework.

Competitor's New Product Development Cycle = 48 months



Results in decreased quality and more rework.

Plan Implement Rework

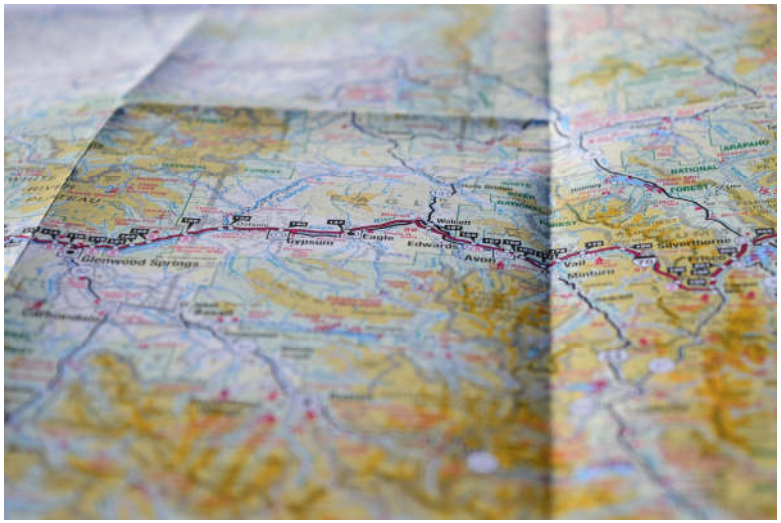
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Strategic Planning (STRAP)

- Your strategy must answer the question…

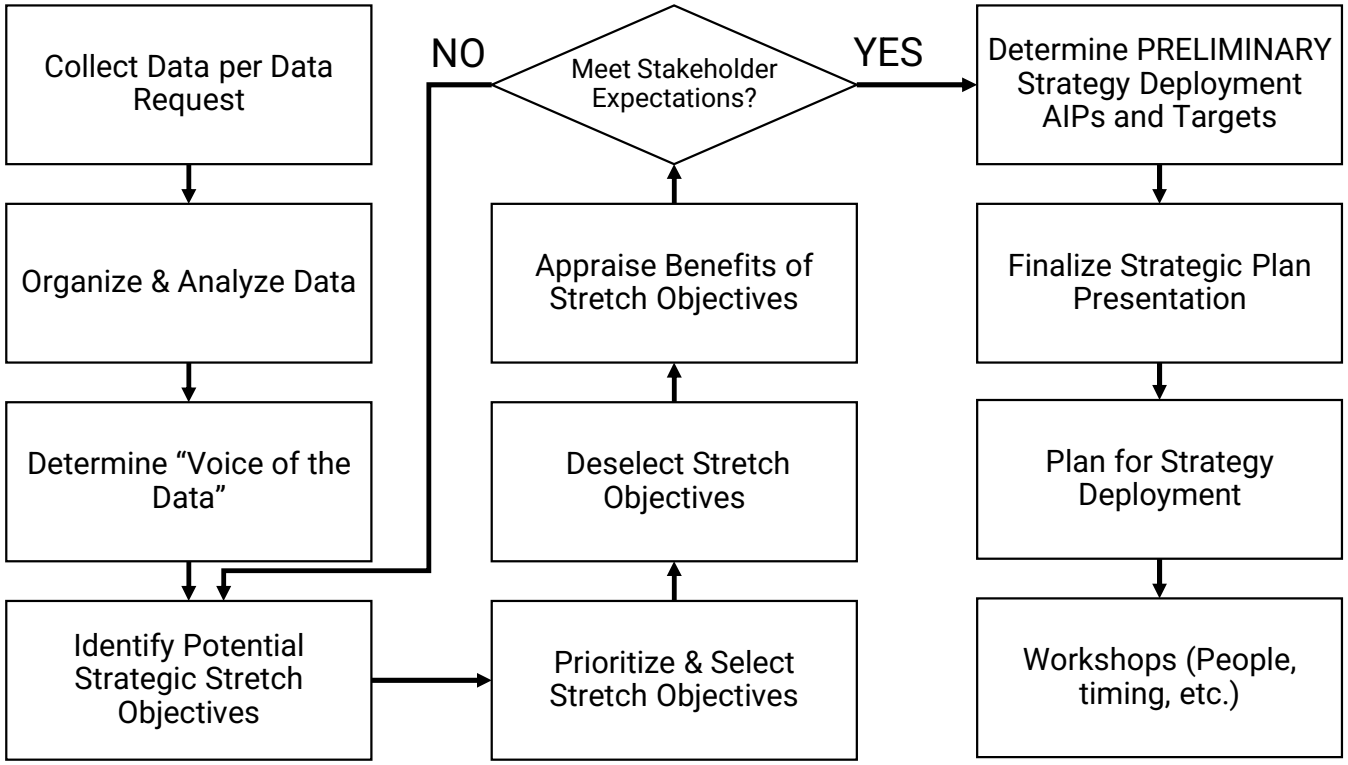
"What game are we playing, and how are we going to win?"

- Winning is defined by your key stakeholders
  - All stakeholders must win!
- The quality of your strategic plan is inversely related to the thickness of the book



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Strategic Planning Roadmap



Strategy is a process!

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Major Strategic Plan Components

1	Mission Statement
2	Vision Statement
3	Major Changes from Last Year
4	Voice of the Customer
5	SWOT Analysis
6	3-Year Sales History
7	Strategic Breakthrough Objectives

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Supporting Data and Analysis

8	Operations & Supply Chain Analysis
9	Market Analysis
10	Competitive Analysis
11	Technological Trends
12	SKU Analysis
13	Risk Analysis
14	Human Capital
15	Financial Summary

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Strategy Deployment

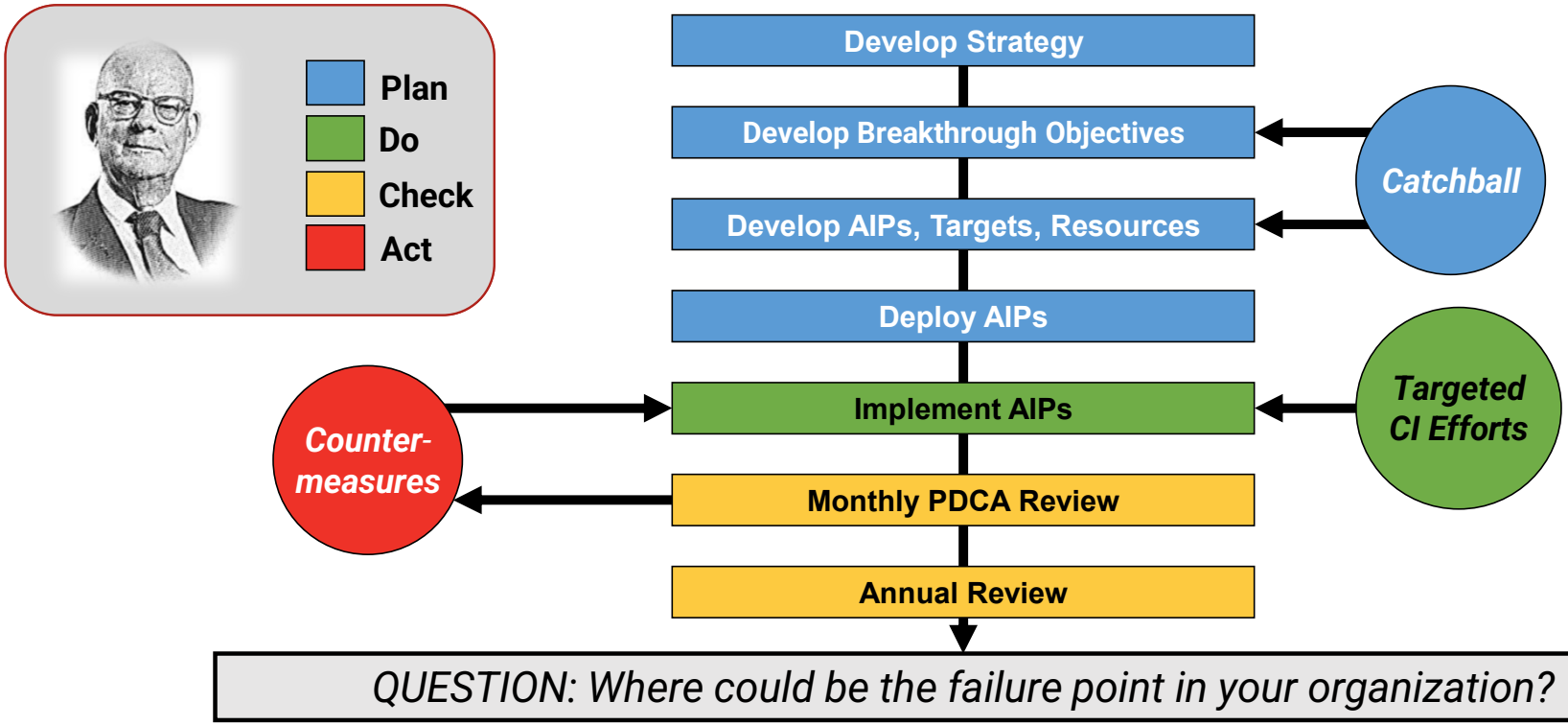
- A logical, three-year plan with one-year components
- Drives breakthrough performance levels
- Aligns valuable resources within the company
- “Operationalizes” strategy by linking to tactics
- Integrated performance review process (PDCA)
- Results in specific action plans—Not discussion
- Facilitates communication of the right vision



Deciding on what NOT to do is as important as deciding what to do!

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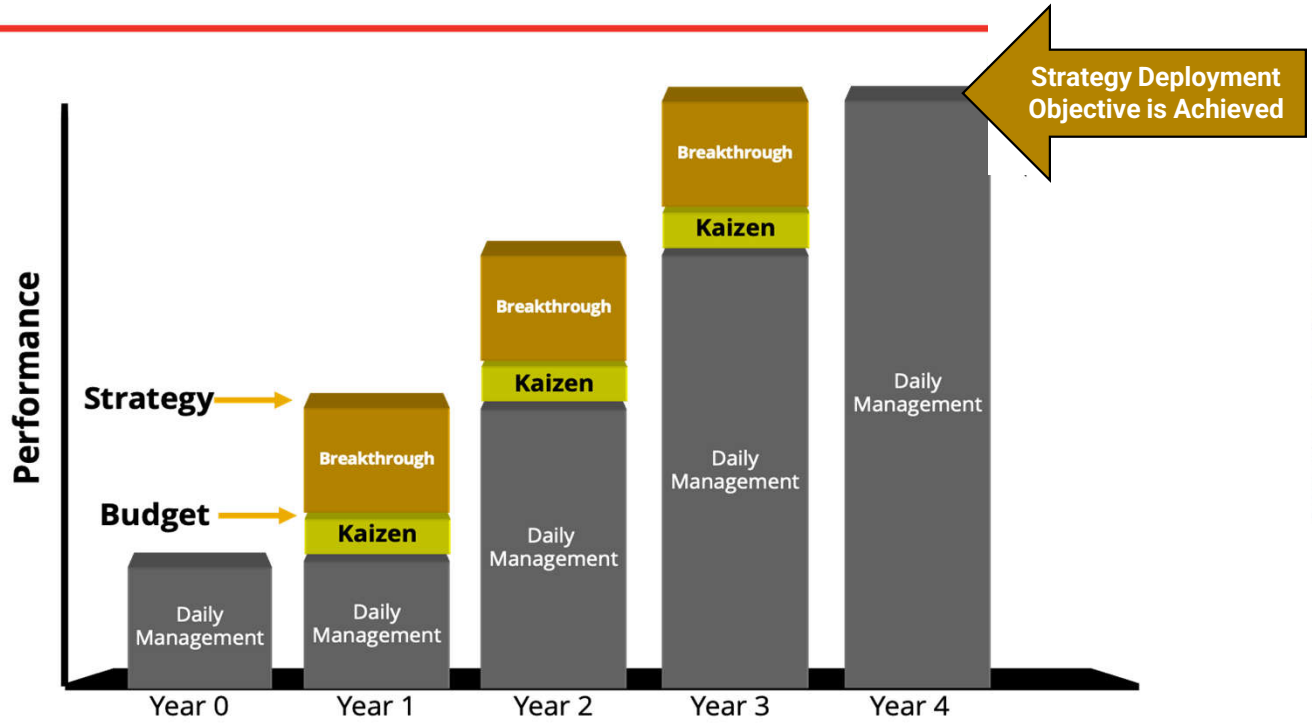
7 Phases of the Strategy Deployment Process Based on the Deming Cycle - PDCA



QUESTION: Where could be the failure point in your organization?

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Breakthrough Thinking





"I believe this nation should challenge itself to put a man on the moon, and return him safely to earth by the end of this decade."

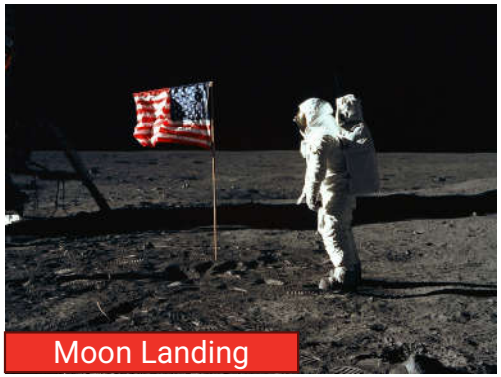
- John F. Kennedy, 1961

QUESTION: Are you using SD to achieve your financial budget or BREAKTHROUGH performance?

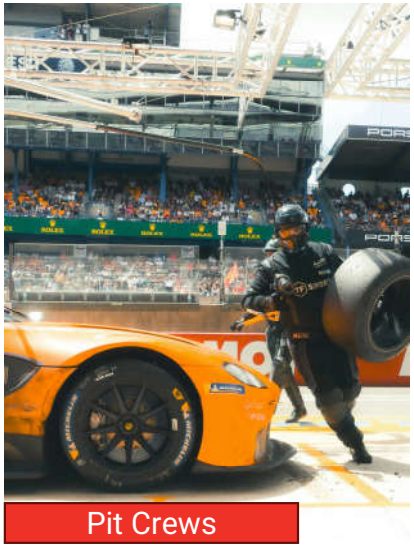
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Breakthrough Examples


Roger Bannister


Moon Landing


Dolly the Sheep


Pit Crews


Summiting Everest


Wright Brothers

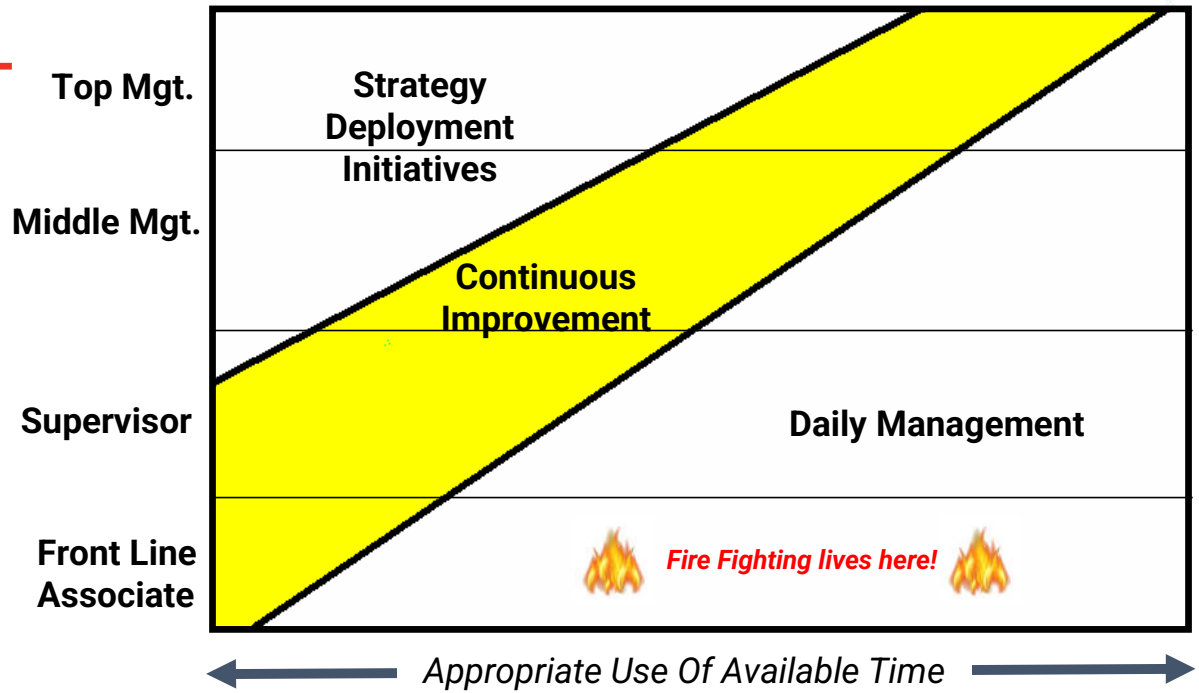
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The Role of Leadership in a Strategy Deployment Environment

Not everyone gets to work on Strategy Deployment.

Someone has to run the business!

But everyone gets to do Continuous Improvement!



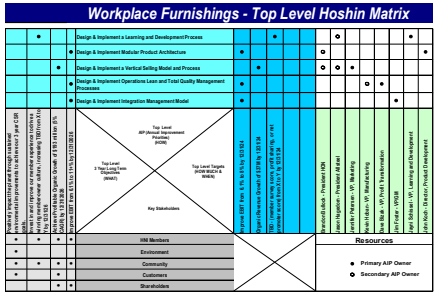
QUESTION: Are you working IN the business, or are you working ON the business?

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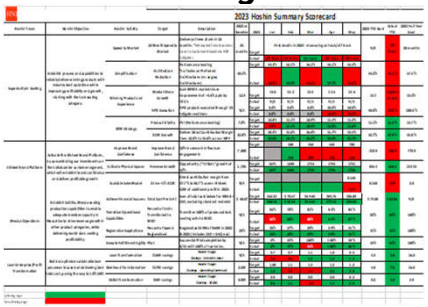
The Forms of Strategy Deployment

PLAN

SD Matrix



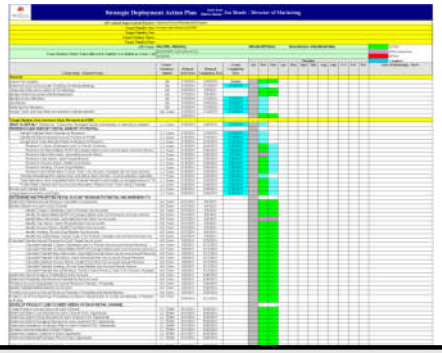
SD Bowling Chart



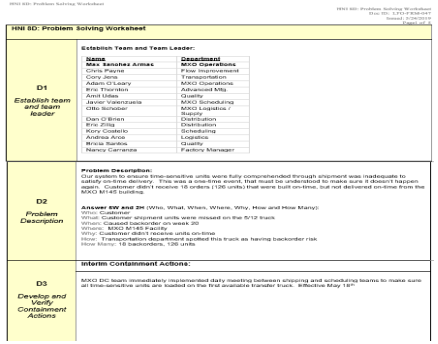
CHECK

DO

Action Plan



8D - Countermeasure Form



ACT

QUESTION: Is your Strategy Deployment a "Form-Filling" Exercise?

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Budgets and Strategy Deployment

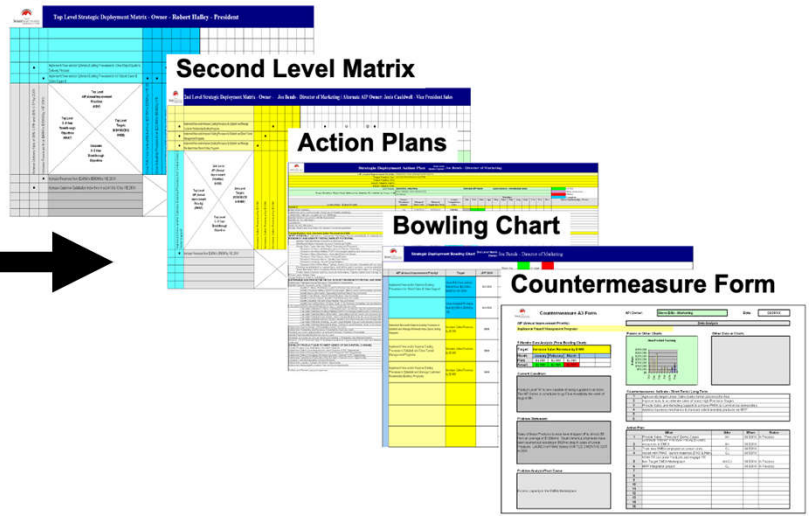
Do Not Use Strategy Deployment to Implement Your Budget

BUDGET/FINANCIAL PLAN



Strategy Deployment

Top Level Matrix

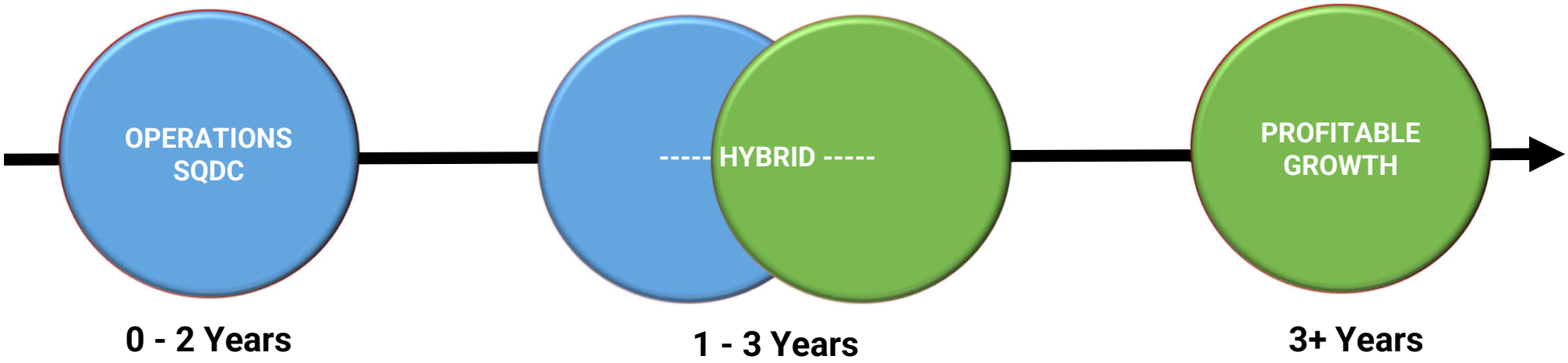


Budgets are NOT Breakthrough!

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The Complexion of Strategy Deployment

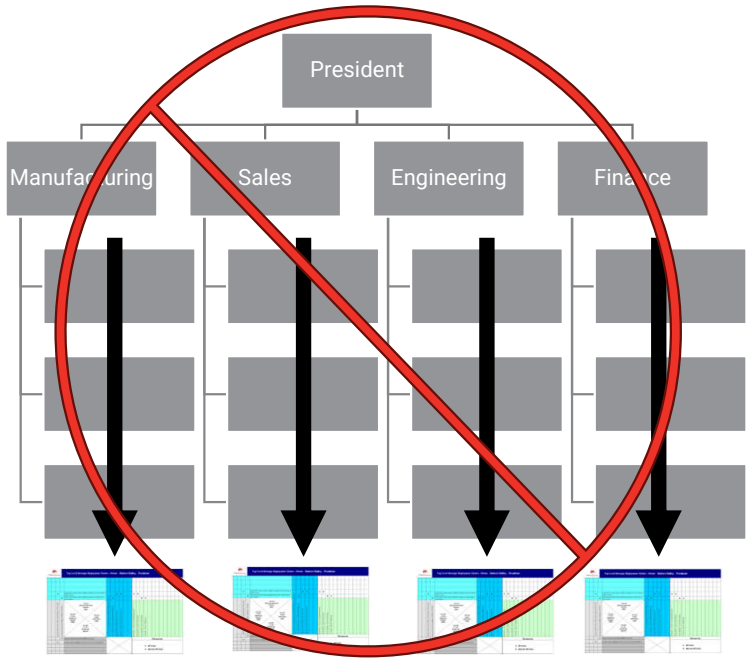
Typical Maturity Timeline - With Aggressive Adoption of Lean

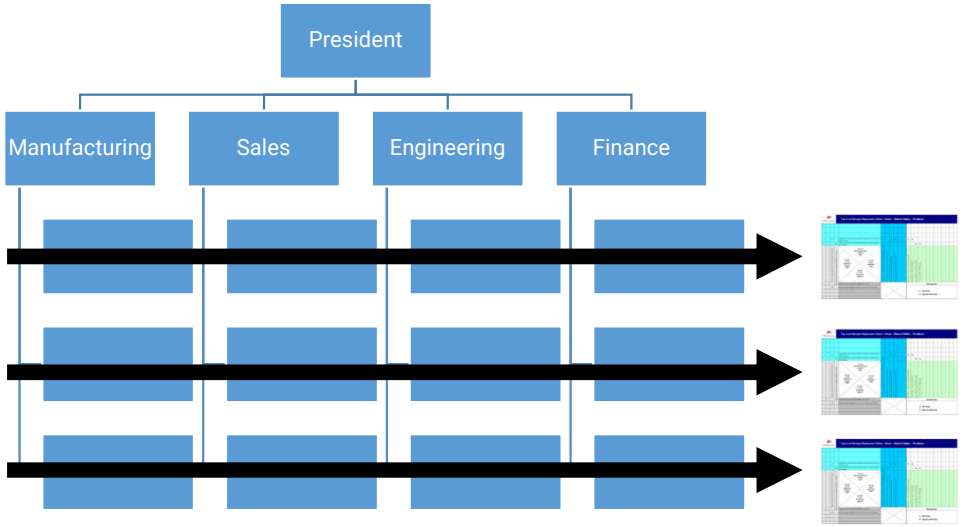


You must earn the right to grow!

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Strategy Deployment: Must be Deployed on a Cross Functional Basis





If you seek to optimize any given function, you will suboptimize the enterprise.

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Daily Management vs. Strategic Deployment

Daily Management

- A way to monitor, manage and improve the daily metrics for each function in the organization.
- Uses the kaizen method to continually improve functional processes.
- Touches everyone in the organization.



Strategic Deployment

- Methodology to implement your few key strategic business imperatives.
- Done by creating vital business processes that will sustainably perform at *Breakthrough* levels.
- Not everyone in the organization will be involved in the implementation.



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Daily Management vs. Strategic Deployment

Criteria	Strategy Deployment	Daily Management
Focus	ON the business	IN the business
People/Teams	Multi-Functional	Functional
Initiatives	Strategic	Tactical
Scope	Narrow & Deep	Wide/Across Enterprise
Metrics	Few - Strategic	Many - KPIs
Improvement	Breakthrough	Incremental
Financial	Beyond Budget	Budget
Involvement	Select Few	All
Kaizen Resource Allocation	70%	30%
Review Cycle	Weekly/Monthly	Hourly/Daily
Management Focus	Senior	Middle & Below

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Daily Management vs. Strategic Deployment

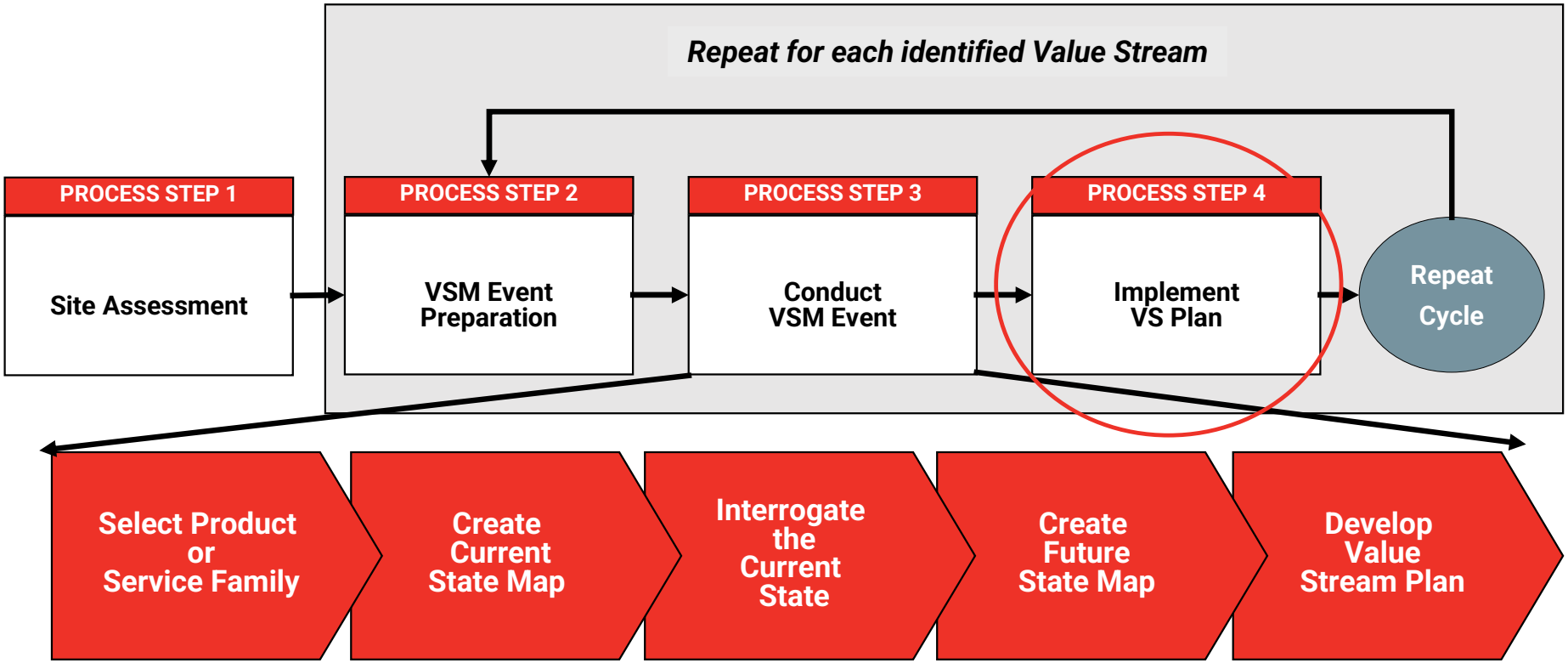
Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. How does your company develop and deploy strategy?
  2. Based on what you've learned, are there areas to improve?
  3. How does your company differentiate between Daily Management and Strategy Deployment?



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Value Stream Mapping Process



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Value Stream Planning

Value Stream Plan

Product Family	
Value Stream	
Current Date	
Start Date	

Business Objective	
Value Stream Manager	
Operations Manager	

Signatures	
CEO	
V.S. Mgr	
Ops Mgr.	

VS Loop	#	Value Stream Objective	Goal (measureable)	Monthly Schedule J	F	M	A	M	J	J	A	S	O	N	D	Target Date	Person in charge	Status

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Completed Value Stream Plan

Value Stream Plan

Product Family	Metal Enclosures
Value Stream	Ed's Product Line
Current Date	16-Sep-06
Start Date	July 10, 2006

Business Objective	Reduce inventory and lead time
Value Stream Manager	Bill Patient
Operations Manager	David Livery

Signatures	
CEO	J.M. Toledo
V.S. Mgr	B. Patient
Ops Mgr.	D. Livery

VS Loop	#	Value Stream Objective	Goal (measureable)	Monthly Schedule J	A	S	O	N	D	J	F	M	A	M	J	Target Date	Person in charge	Status
Supplier Loop	1	Create Pull with Supplier	Reduce Raw material Inventory from 8.5 days to 6 days													09/01/06	H. Baker	Complete
Mfg Loop	2	Reduce Manpower	Reduce # of operators/machines in CUT													09/01/06	J. Moreno	Complete
	3	Link Stamp & Brake Form Operations in Continuous Flow	Reduce Inventory between operations from 6.2 days to 0 days													10/31/06	F.Baskins	Pending
	4	Reduce Cycle time of Brake Form operation	Reduce DOR from 20 sec to 15 sec													12/01/06	F. Baskins	Pending
	5	Improve Quality of Brake Form operation	Improve first time yield from 85% to 98%													12/01/06	F. Baskins	Pending
	6	Create Pull from Shipping to Stamp/Brake Form to	Reduce Inventory between operations from 5 days to 4 days													12/31/06	H. Baker	Pending
Information Loop	7	Eliminate weekly MRP build scheduling	Mfg operations are triggered by kanban													12/31/06	R. Bonds	Pending
	8	Initiate 30/60/90 day forecast from customer	Increase accuracy of forecast planning by 50%													01/15/07	R. Bonds	Pending

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Site Plan

A single plan and schedule of all kaizen and continuous improvement events for a particular facility.

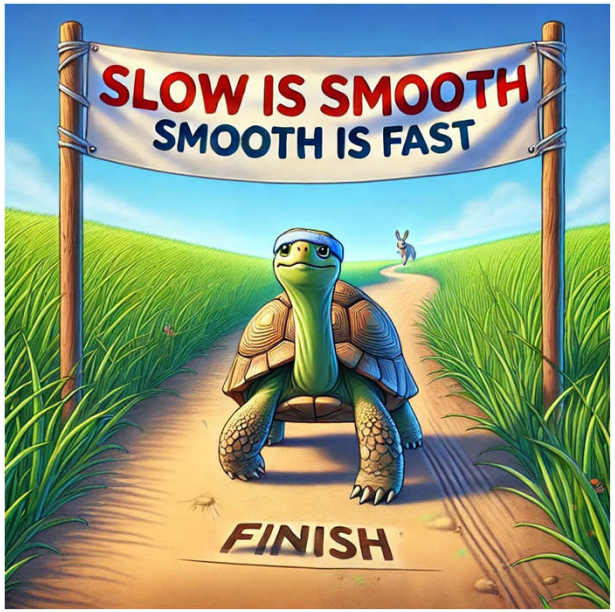
Sample Site Plan excerpt showing scheduled events from two different Value Streams (KVF and SMMM)

WEEK	AREA/CELL	VSM	TOOL	VS Plan Number	STATUS	SENSEI	COMMENTS
3-Jan	**LINE 8 (KVF)**	**KVF**	**Standard work**	**3**	**complete**	**Clem**	**see kaizen results sheet**
3-Jan	**LINE 25**	**SMMM**	**Standard Work**	**8**	**complete**	**Clem**	**see kaizen results sheet**
10-Jan	**ATJ**	**KVF**	**SMED**	**4**	**complete**	**Clem**	**PRE-RUN S/U WHEEL**
10-Jan	**Anodize racking**	**SMMM**	**Standard work**	**6**	**complete**	**Clem**	**see kaizen results sheet**
17-Jan	**NO EVENTS**						**Clem out**
24-Jan	**ATJ**	**KVF**	**Standard Work**	**2**	**complete**	**Clem**	**DEMAND QUESTIONS**
24-Jan	**LINE 28 (KVF new area)**	**KVF**	**Standard work**	**1**	**complete**	**Clem**	**DEMAND QUESTIONS**
24-Jan	**LINE 25**	**SMMM**	**5S**	**3**	**complete**	**Clem**	**CELL MEMBERS ONLY**

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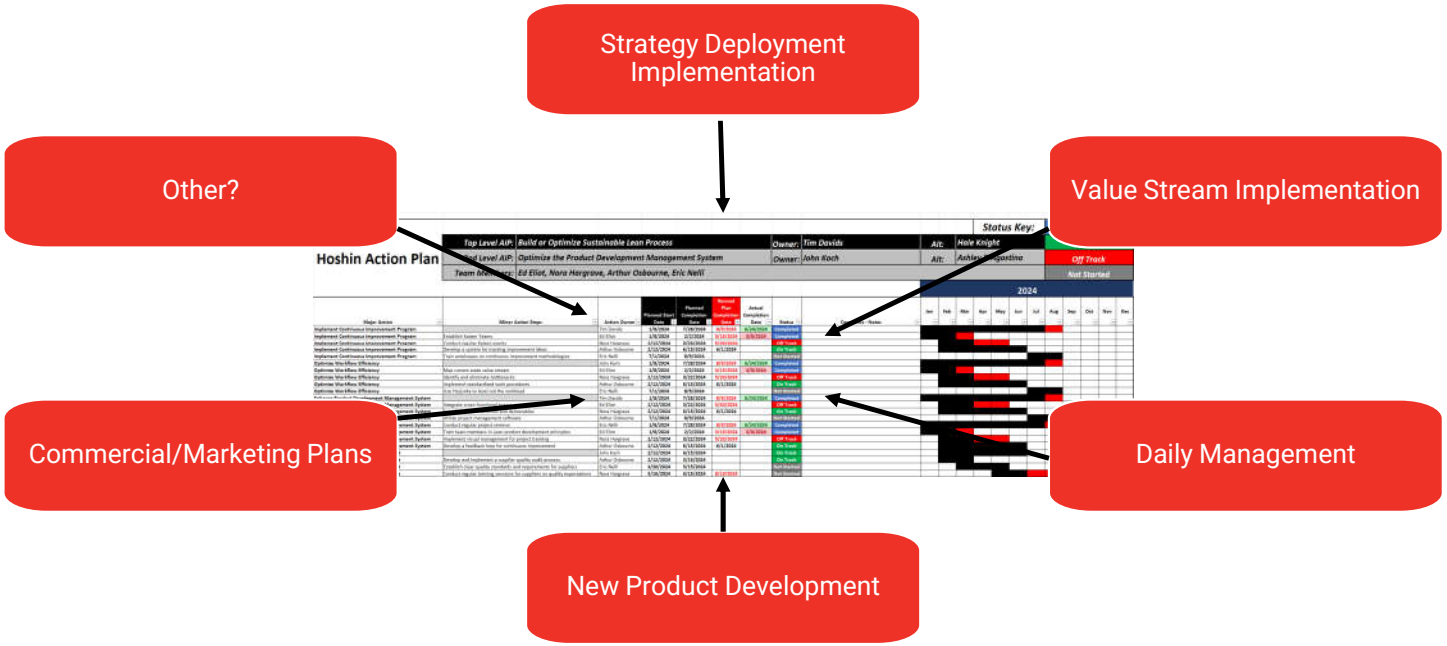
Action Planning

- Action Plans are a detailed list of actions and tasks necessary for the successful implementation of a project, strategy deployment, value stream, etc.
- Action Plans are the “recipes” used to create a process/system.
- Every action plan should have, at a minimum:
  - **WHAT** – The task or activity
  - **WHO** – The person or team tasked to complete the activity
  - **WHEN** – The date by which a task/activity should be completed
  - **MILESTONE** – Points along the way that represent some achievement towards progress



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Possible Action Plan Uses



Any Other Project Requiring Management of Resources, Timelines, and Activities

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Action Planning

Example

Status Key:	
	Completed
	On Track
	Off Track
	Not Started

Top Level AIP:	Build or Optimize Sustainable Lean Process	Owner:	Tim Davids	Alt:	Hale Knight
2nd Level AIP:	Optimize the Product Development Management System	Owner:	John Koch	Alt:	Ashley D'Agostino
Team Members:	Ed Eliot, Nora Hargrave, Arthur Osbourne, Eric Nelli				

Major Action	Minor Action Steps	Action Owner	Planned Start Date	Planned Completion Date	Revised Plan Completion Date	Actual Completion Date	Status	Comments - Notes
Implement Continuous Improvement Program		Tim Davids	1/8/2024	7/28/2024	8/9/2024	6/24/2024	Completed	
Implement Continuous Improvement Program	Establish Kaizen Teams	Ed Eliot	1/8/2024	2/2/2024	3/15/2024	3/8/2024	Completed	
Implement Continuous Improvement Program	Conduct regular Kaizen events	Nora Hargrave	2/12/2024	3/23/2024	5/20/2024		Off Track	
Implement Continuous Improvement Program	Develop a system for tracking improvement ideas	Arthur Osbourne	2/12/2024	6/15/2024	6/1/2024		On Track	
Implement Continuous Improvement Program	Train employees on continuous improvement methodologies	Eric Nelli	7/1/2024	8/9/2024			Not Started	
Optimize Workflow Efficiency		John Koch	1/8/2024	7/28/2024	8/9/2024	6/24/2024	Completed	
Optimize Workflow Efficiency	Map current state value stream	Ed Eliot	1/8/2024	2/2/2024	3/15/2024	3/8/2024	Completed	
Optimize Workflow Efficiency	Identify and eliminate bottlenecks	Nora Hargrave	2/12/2024	3/22/2024	5/20/2024		Off Track	
Optimize Workflow Efficiency	Implement standardized work procedures	Arthur Osbourne	2/12/2024	6/15/2024	6/1/2024		On Track	
Optimize Workflow Efficiency	Use Heijunka to level out the workload	Eric Nelli	7/1/2024	8/9/2024			Not Started	
Enhance Product Development Management System		Tim Davids	1/8/2024	7/28/2024	8/9/2024	6/24/2024	Completed	
Enhance Product Development Management System	Integrate cross-functional teams	Ed Eliot	2/12/2024	3/22/2024	5/20/2024		Off Track	
Enhance Product Development Management System	Establish clear milestones and deliverables	Nora Hargrave	2/12/2024	6/15/2024	6/1/2024		On Track	
Enhance Product Development Management System	Utilize project management software	Arthur Osbourne	7/1/2024	8/9/2024			Not Started	
Enhance Product Development Management System	Conduct regular project reviews	Eric Nelli	1/8/2024	7/28/2024	8/9/2024	6/24/2024	Completed	
Enhance Product Development Management System	Train team members in Lean product development principles	Ed Eliot	1/8/2024	2/2/2024	3/15/2024	3/8/2024	Completed	
Enhance Product Development Management System	Implement visual management for project tracking	Nora Hargrave	2/12/2024	3/22/2024	5/20/2024		Off Track	
Enhance Product Development Management System	Develop a feedback loop for continuous improvement	Arthur Osbourne	2/12/2024	6/15/2024	6/1/2024		On Track	
Improve Supplier Quality Management		John Koch	2/12/2024	6/15/2024			On Track	
Improve Supplier Quality Management	Develop and implement a supplier quality audit process	Arthur Osbourne	2/12/2024	3/15/2024			On Track	
Improve Supplier Quality Management	Establish clear quality standards and requirements for suppliers	Eric Nelli	3/20/2024	5/15/2024			Not Started	
Improve Supplier Quality Management	Conduct regular training sessions for suppliers on quality expectations	Nora Hargrave	5/16/2024	6/15/2024	8/15/2024		Not Started	

2024

Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec

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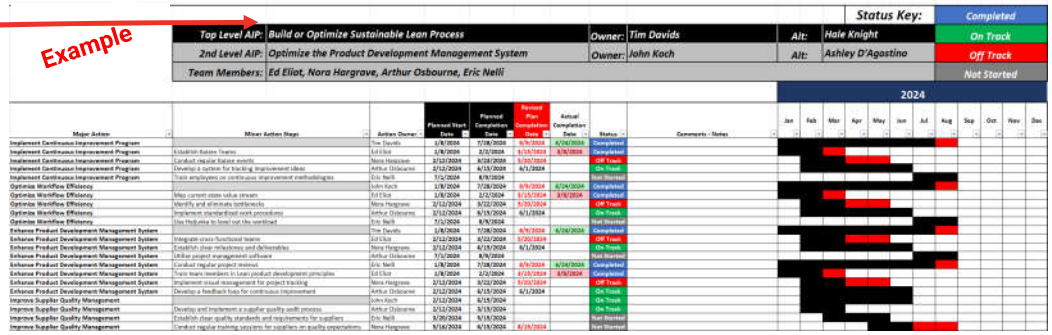
How to Use Action Planning

Value Stream Plan

Product Family	Metal Enclosures	Business Objective	Reduce inventory and lead time	Signatures: CEO	T.M. Davids
Value Stream	Ed's Product Line	Value Stream Manager	Bill Patient	V.S. Mgr	B. Patient
Current Date	16-Sep-06	Operations Manager	David Livery	Ops Mgr.	D. Livery
Start Date	July 10, 2006				

VS Loop	#	Value Stream Objective	Goal (measureable)	Monthly Schedule J	A	S	O	N	D	J	F	M	A	M	J	Target Date	Person in charge	Status
Supplier Loop	1	Create Pull with Supplier	Reduce Raw material inventory from 8.5 days to 6 days													09/01/06	H. Baker	Complete
Mfg Loop	2	Reduce Manpower	Reduce # of operators/machines in CUT													09/01/06	J. Moreno	Complete
	3	Link Stamp & Brake Form Operations in Continuous Flow	Reduce Inventory between operations from 6.2 days to 0 days													10/31/06	F. Baskins	Pending
	4	Reduce Cycle time of Brake Form operation	Reduce DOR from 20 sec to 15 sec													12/01/06	F. Baskins	Pending
	5	Improve Quality of Brake Form operation	Improve first time yield from 85% to 98%													12/01/06	F. Baskins	Pending
	6	Create Pull from Shipping to Stamp/Brake Form to	Reduce Inventory between operations from 5 days to 4 days													12/31/06	H. Baker	Pending
Information Loop	7	Eliminate weekly MRP build scheduling	Mfg operations are triggered by kanban													12/31/06	R. Bonds	Pending
	8	Initiate 30/60/90 day forecast from customer	Increase accuracy of forecast planning by 50%													01/15/07	R. Bonds	Pending

Example



30,000 Foot View

Ground Level

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Resource Collision Planning

2nd Level AIP Manpower Workload Analysis

Top Level AIP: ***Design & Implement Modular Product Architecture***

Name	AIP 1		AIP 2		AIP 3		AIP 4		AIP 5		Total Load %
	● or ❍	%	● or ❍	%	● or ❍	%	● or ❍	%	● or ❍	%	
Bob	●	25%							❍	10%	35%
Mary	❍	10%									10%
Susan			❍	20%	❍	15%	❍	15%			50%
Richard									❍	15%	15%
Charlie	❍	20%	●	25%	●	30%	❍	40%	❍	10%	125%
Sam	❍	15%									15%
Jennifer			❍	15%							15%
Todd					❍	20%	●	30%			50%
Mark	❍	50%									50%
Jeff			❍	10%	❍	10%			❍	20%	40%
Billy							❍	20%			20%
Tommy					❍	25%					25%
Leslie							❍	15%			15%
Leigh	❍	20%	❍	50%			❍	10%	●	25%	105%
Andrea					❍	30%					30%
Diane									❍	10%	10%
Robert			❍	20%							20%
James	❍	30%			❍	10%	❍	25%			65%
Total Load %	170%		140%		140%		155%		90%		
Equiv FT Heads	1.7		1.4		1.4		1.6		0.9		7.0

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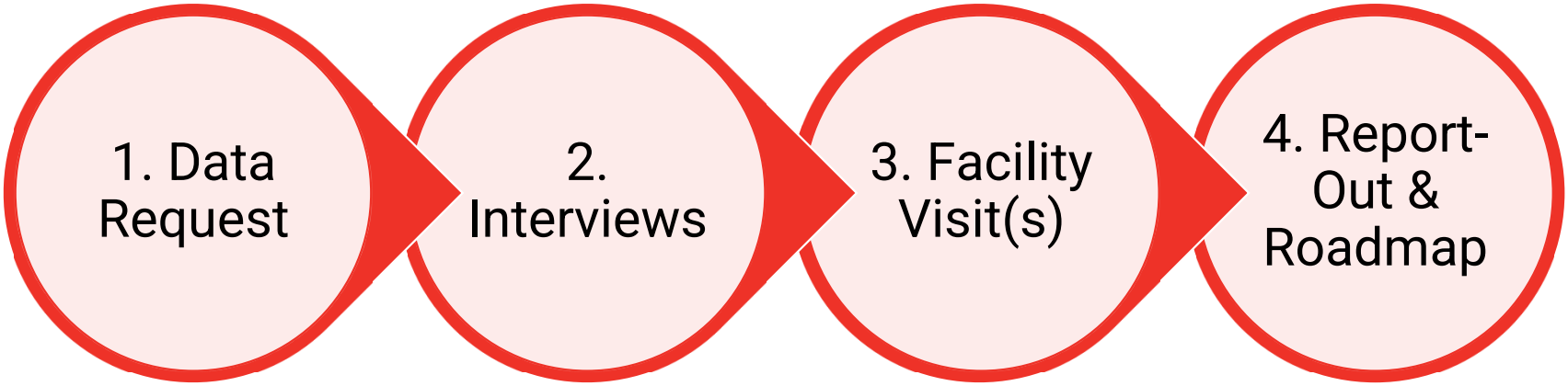
Assessments:
Where We Are -> Where We're Going

- Using an assessment process (3rd party -> internal) to evaluate the business.
- Checkpoint in time on overall Lean transformation and effectiveness.
- An unbiased evaluation of effort and progress.
- A muscle your Lean team needs to develop.



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Assessment Roadmap



- Business Strategy
- Financial Data
- Organizational Data
- Operating Data
- Lean Office Data

- Executive Leadership
- Management
- Supervisory
- Front Line

- Observations - Guided
- Observations - Independent
- Interviews

- *To be presented at the Lean Boot Camp*
- What We Heard vs. What We Observed
- Draft Implementation Roadmap
- Operational Benefits
- Financial Benefits w/ROI

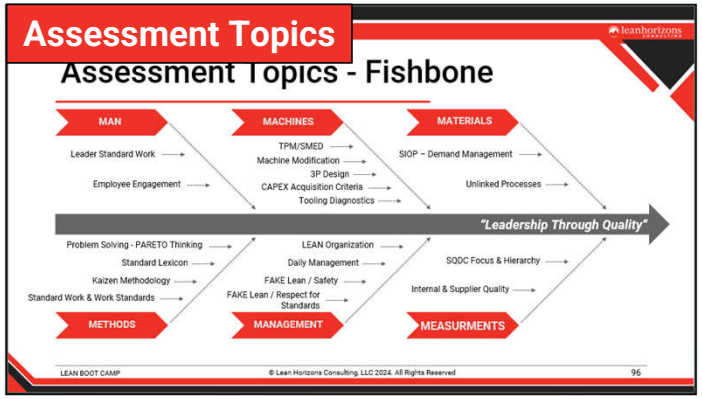
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Sample Slides from a Lean Assessment

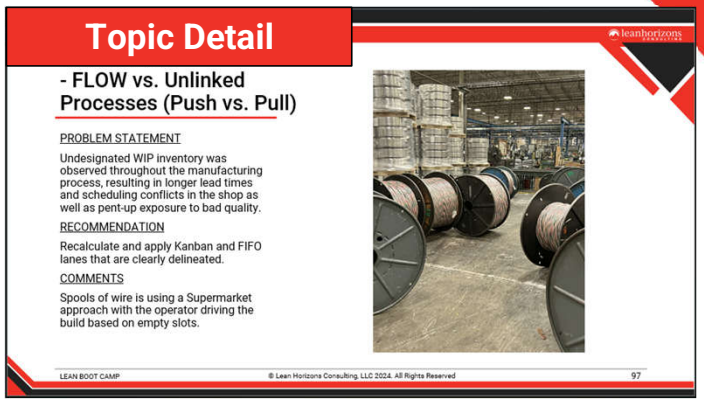
Agenda



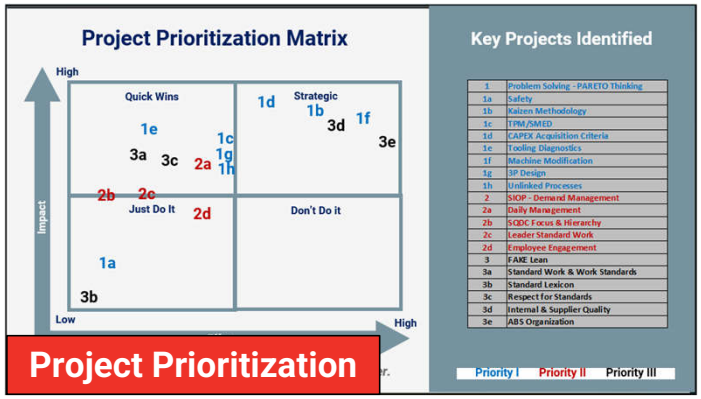
Assessment Topics



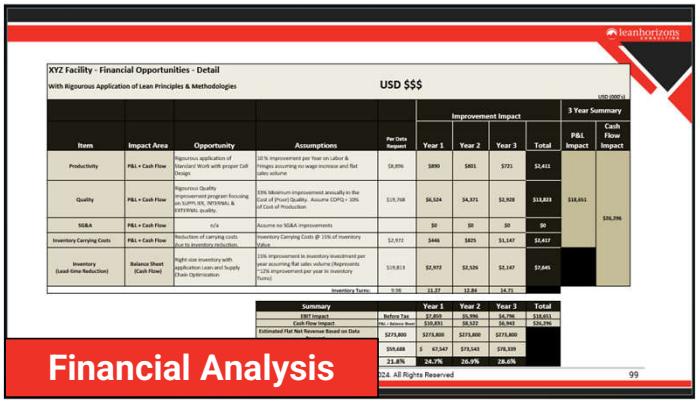
Topic Detail



Project Prioritization



Financial Analysis



See Appendix A for full sample slides

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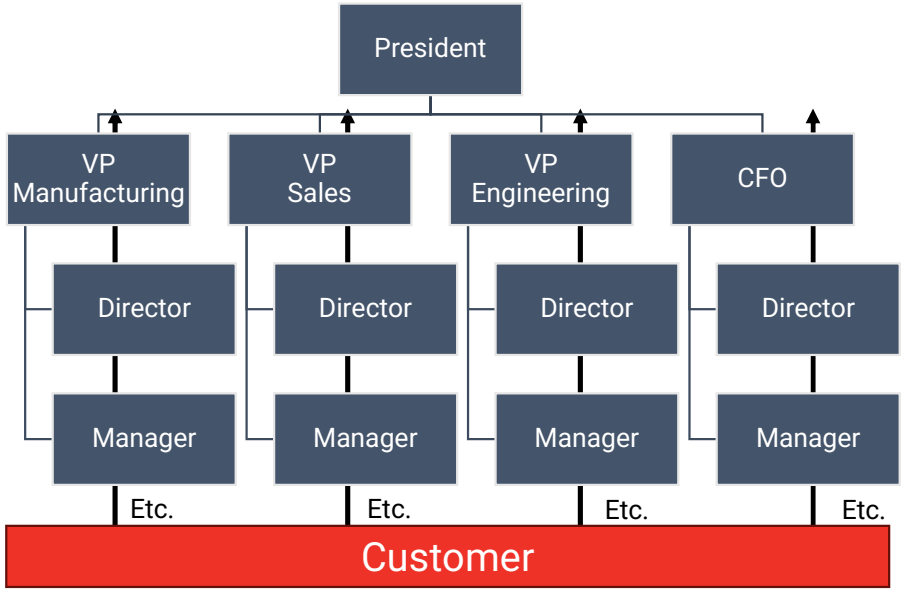
Organizational Planning & Development

Talent Management	
Onboarding Process	Optimize time-to-productivity. Process may vary depending on the role being onboarded.
Succession Planning Process	Identify and document skills and knowledge for future employees.
Career/High-Potential Development Process	Individual career development plans for employees showing potential for advancement and increased responsibilities.
Talent Acquisition and Recruiting Process	Long-term strategic planning for hard-to-fill positions vs. short-term regular hiring for entry-level roles.
Compensation Management Process	Salary, incentives, and benefits each employee receives.
Performance Management Process	Align employee's efforts with the organization's strategic goals.

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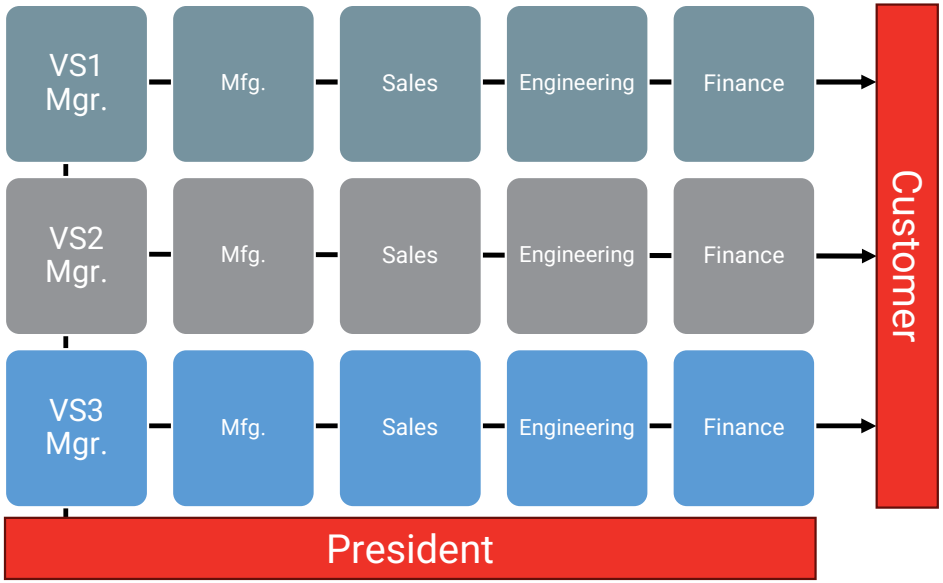
Organizational Planning & Development

Functional Structure



The customer's needs are subordinate to functional leaders.

Value Stream Structure



The customer's needs are what drive all actions in the business.

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Organizational Planning & Development

How would a value stream structure work in your organization?



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Organizational Planning & Development

Breakout Discussion:

- Discuss the following questions and be prepared to present your answers:
  1. How would you develop a value stream organization in your company?
  2. Would your value stream be structured around:
     a. Product family?
        i. Size/Dimension?
        ii. Color?
        iii. MTO vs. MTS, etc.?
     b. Process?
     c. Customer?
     d. Geography?
     e. Other/Combination of the above?



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Plan

- Strategic Planning (STRAP)
- Strategy Deployment
- Value Stream Planning
- Action Planning
- Resource Collision Planning
- Assessment Roadmap
- Organizational Planning & Development

What were your key takeaways from this section?

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Process

- 10 Rules of a Process
- Enterprise Lean Excellence
- Lean Tools



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Process Defined

A systematic series of activities directed toward achieving an end result in which one or more inputs are acted upon to create one or more outputs.

Applies to products, services, information, etc.



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Process Rule #1

If a process is not documented, it is not a process!

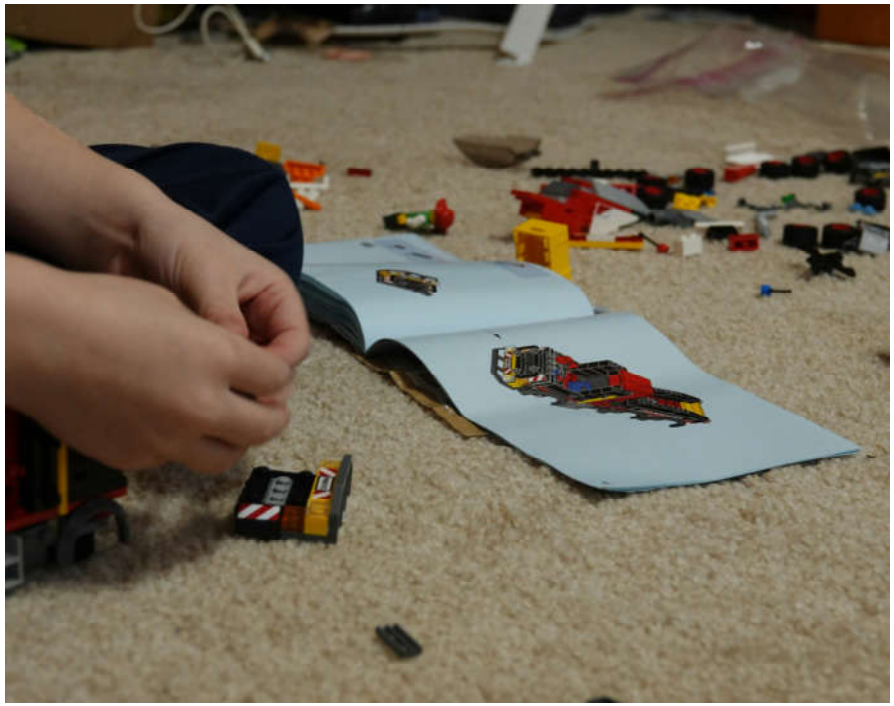


Value Stream Mapping, Process Mapping, Standard Work Documentation, Job Instructions, Process at a Glance, etc.

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Process Rule #2

Processes should be displayed visually, where applicable.



Visual Management, U-Shaped Cells, GEMBA Boards, Value Stream Mapping, Process Mapping, Flow Charts, etc.

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Visual Management Lego Exercise

Instructions:

- Choose two individuals to build the Lego sets.
- Choose two individuals to serve as timekeepers for the builders.
- Supplies for builders:
  - A set of Lego bricks.
  - One builder gets visual instructions.
  - The other gets written instructions.
- Record the final times.
- Discuss the results.



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Process Rule #3

Each process step should have detailed job instructions, with assigned roles and responsibilities.

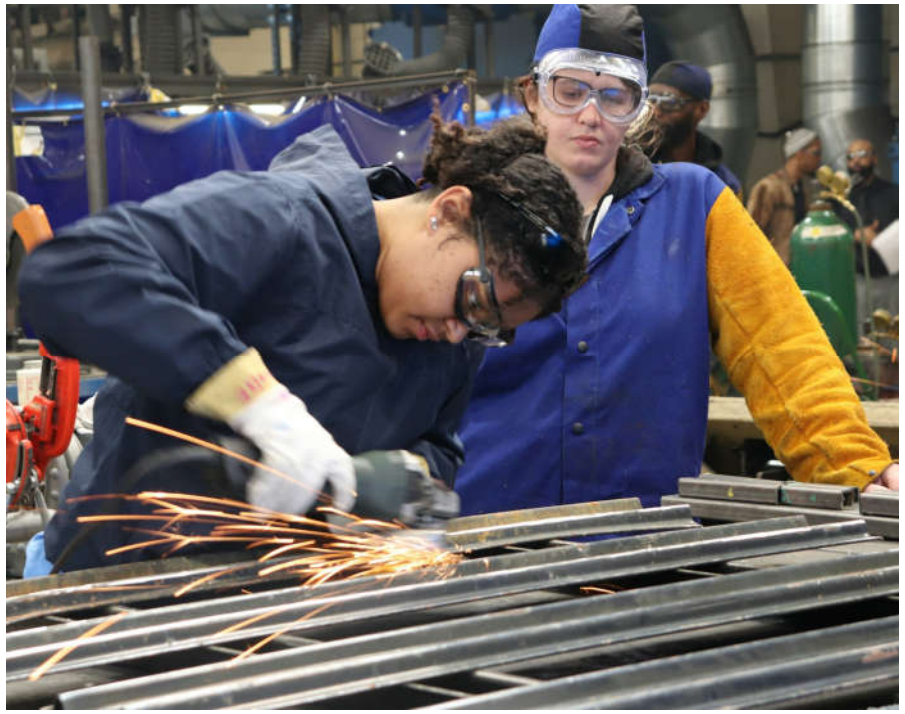


Training Within Industries (TWI)

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Process Rule #4

All associates need to be properly trained on the process, with periodic refresher training.



Kaizen Events, Training Within Industries (TWI)

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Process Rule #5

All movement of a product, document, or information should have a clear and direct path.



Standard Work

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Process Rule #6

Assure the quality of the product or service before passing it on to the next process/customer.



Poka Yoke, Jidoka, One-Piece Flow

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Process Rule #7

WAITING waste is perhaps the most detrimental waste in a process.



FLOW First, PULL where you can't, NEVER PUSH!!!

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Process Rule #8

Processes need to be evolved through rigorous application of Kaizen.



改善

KAIZEN

70% of Kaizen should be directed at Strategic Deployment imperatives, and 30% towards Daily Management

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Process Rule #9

Key Performance Indicators (KPIs) should be developed to track the following:

- Adherence to the process
- Process outcomes

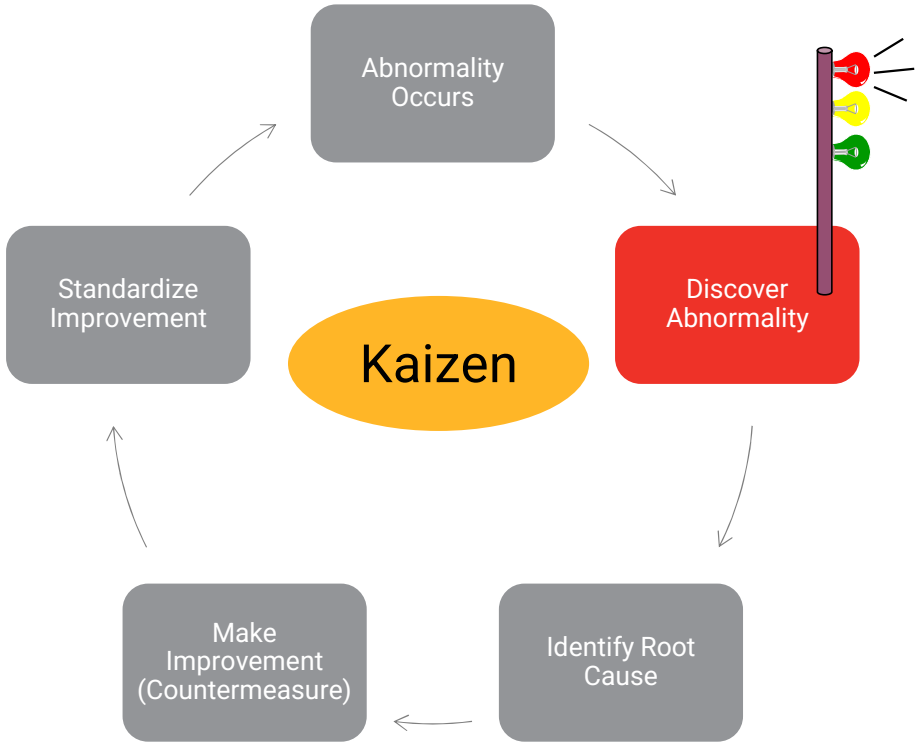


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Process Rule #10

Apply the Deming Cycle (PDCA) to continually monitor and correct deviations from the process.

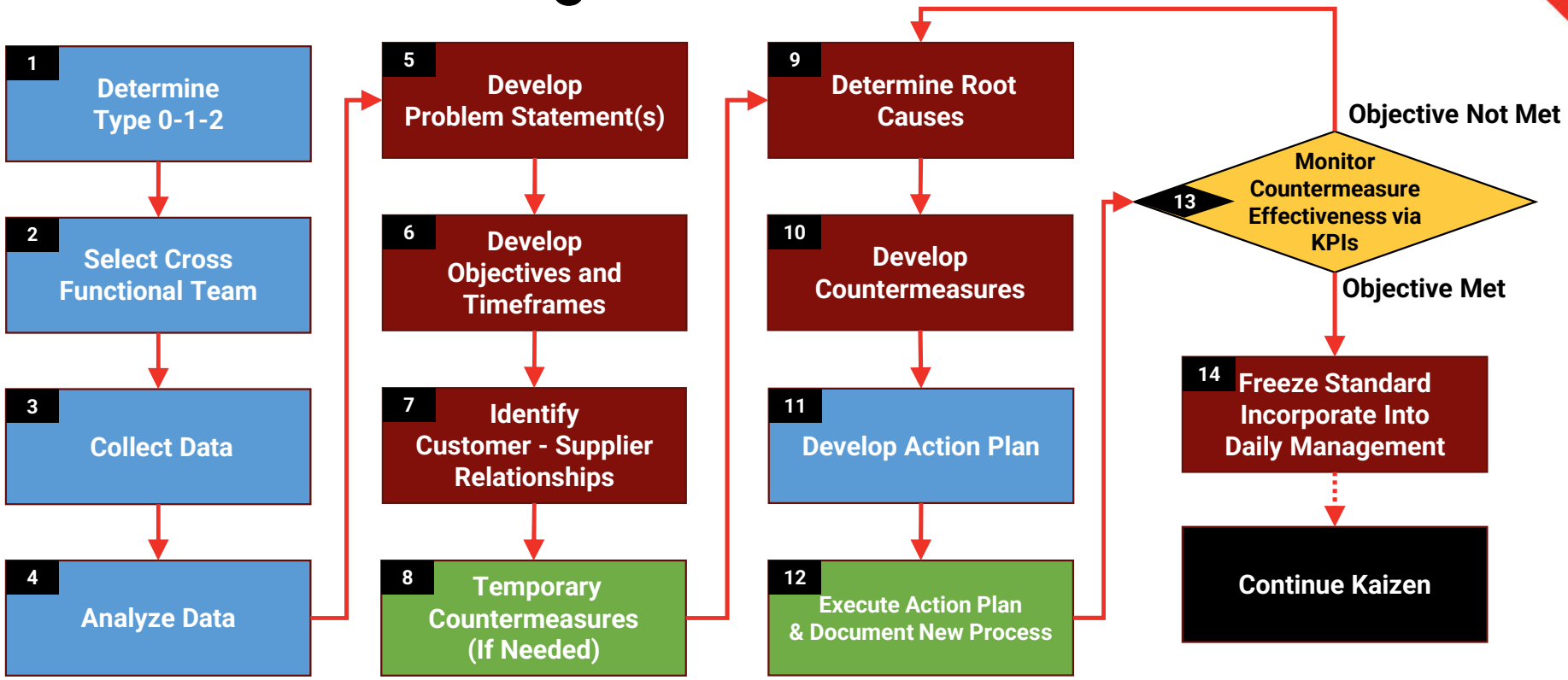




Root Cause Countermeasures (A3), CEDAC, 5 Whys?, other Problem-Solving approaches as needed.

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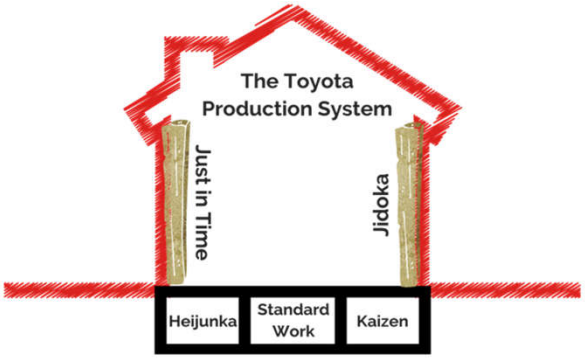
LHC Problem-Solving Process

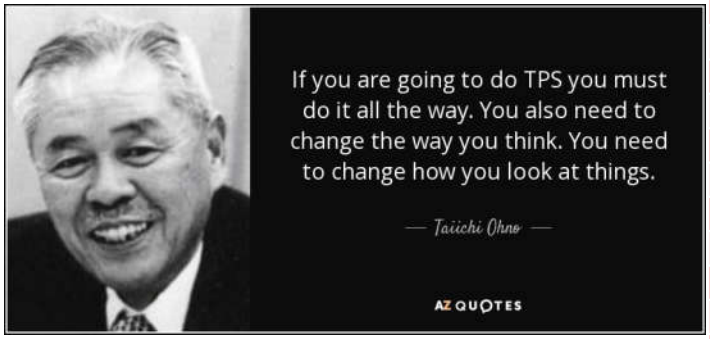


Key: Plan | Do | Check | Act

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Lean Tools





Lean Tools	Mfg	Administration	Growth	Lean Business Office
Strategic Planning (STRAP)	●	●	●	
Strategy Deployment (Hoshin)	●	●	●	
Lean Leadership and Coaching	●	●	●	
Lean Transformation Architecture	●	●	●	
Sales Inventory Operational Planning/Heijunka	●	●	●	
New Product Development	●		●	
3P (Production Preparation Process)/Moonshine	●		●	
Root Cause Problem Solving	●	●	●	
Supplier Development and Certification	●		●	
VOC/QFD	●		●	
Facilitation Skills/Certification	●	●	●	●
Kaizen Standard Work	●	●	●	●
LBO Structure and Organization	●	●	●	●
Introduction to Lean	●	●		
5S	●	●		
Cell/Flow Design	●	●		
Kanban/PFEP	●			
Poka Yoke/Jidoka	●	●		
Single Minute Exchange of Dies (SMED)	●			
Standard Work	●	●		
Total Productive Maintenance (TPM)	●			
Visual Management	●	●		
Value Stream Mapping	●	●		
Lean Accounting	●	●		
Process Mapping	●	●		

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Closing Thought #1

Principles & Values without a supporting process are merely slogans.



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Closing Thought #2

Toyota gets BRILLIANT results with ORDINARY people running BRILLIANT processes.



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Process

- 10 Rules of a Process
- Enterprise Lean Excellence
- Lean Tools

What were your key takeaways from this section?

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Objectives (Recap)

- Introduce and/or reinforce the framework of a Lean transformation.
- Compare and contrast your organization through a Lean lens.
- To stimulate dialogue to discover the compatibility of the Lean framework with your organization.
- Understand your role as a leader in a Lean transformation.
- Develop non-negotiables as they relate to the Lean framework.

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Takeaways

- What are your three biggest takeaways from this Boot Camp?
- Are any areas unclear to you?



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Closing Thought

You are perfectly engineered to get the results you have already gotten.

Want to change the results?



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Agenda

2- Assessment
The 6 - Ms

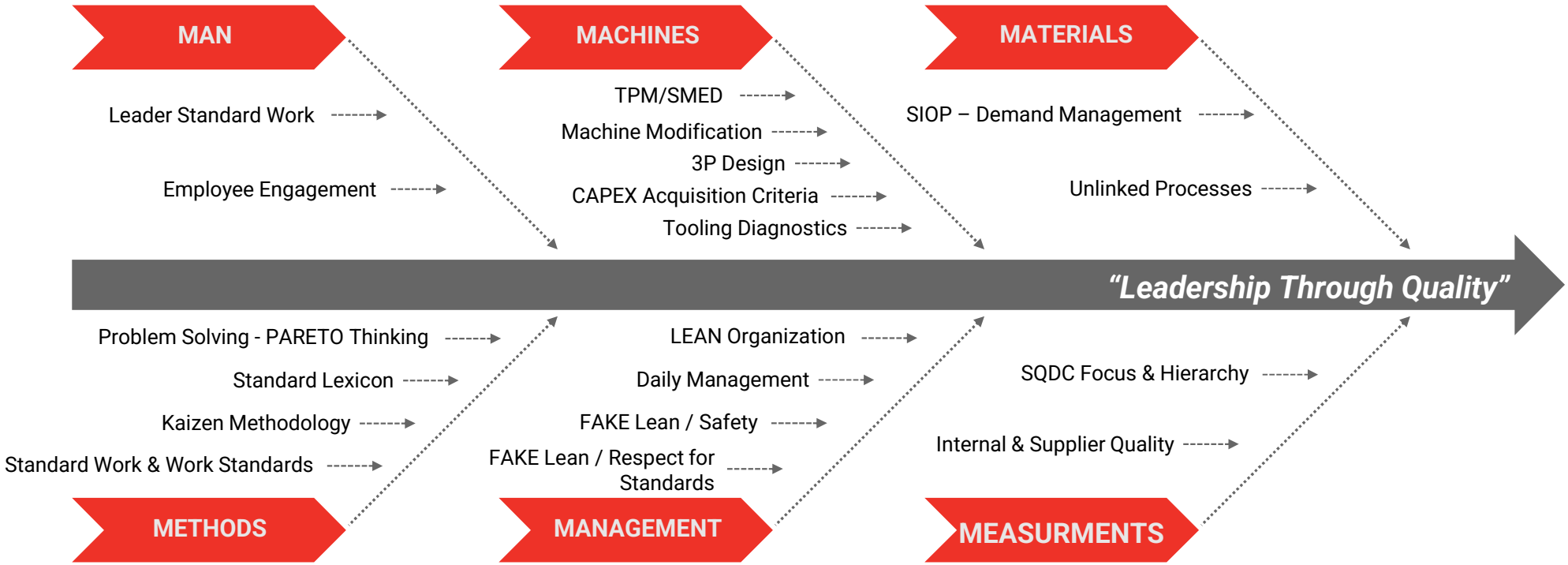
3 - Recommendations
& Go-Forward Plan

1. Introduction

4 - Financial Opportunities

5 - Final Thoughts - Q&A

Assessment Topics - Fishbone



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MATERIALS
- FLOW vs. Unlinked Processes (Push vs. Pull)

PROBLEM STATEMENT

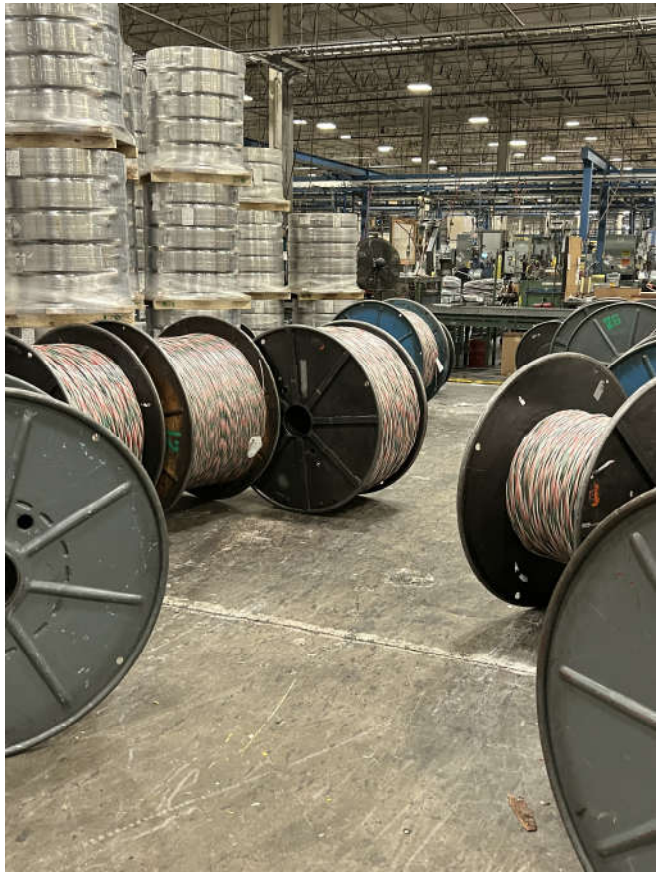
Undesignated WIP inventory was observed throughout the manufacturing process, resulting in longer lead times and scheduling conflicts in the shop as well as pent-up exposure to bad quality.

RECOMMENDATION

Recalculate and apply Kanban and FIFO lanes that are clearly delineated.

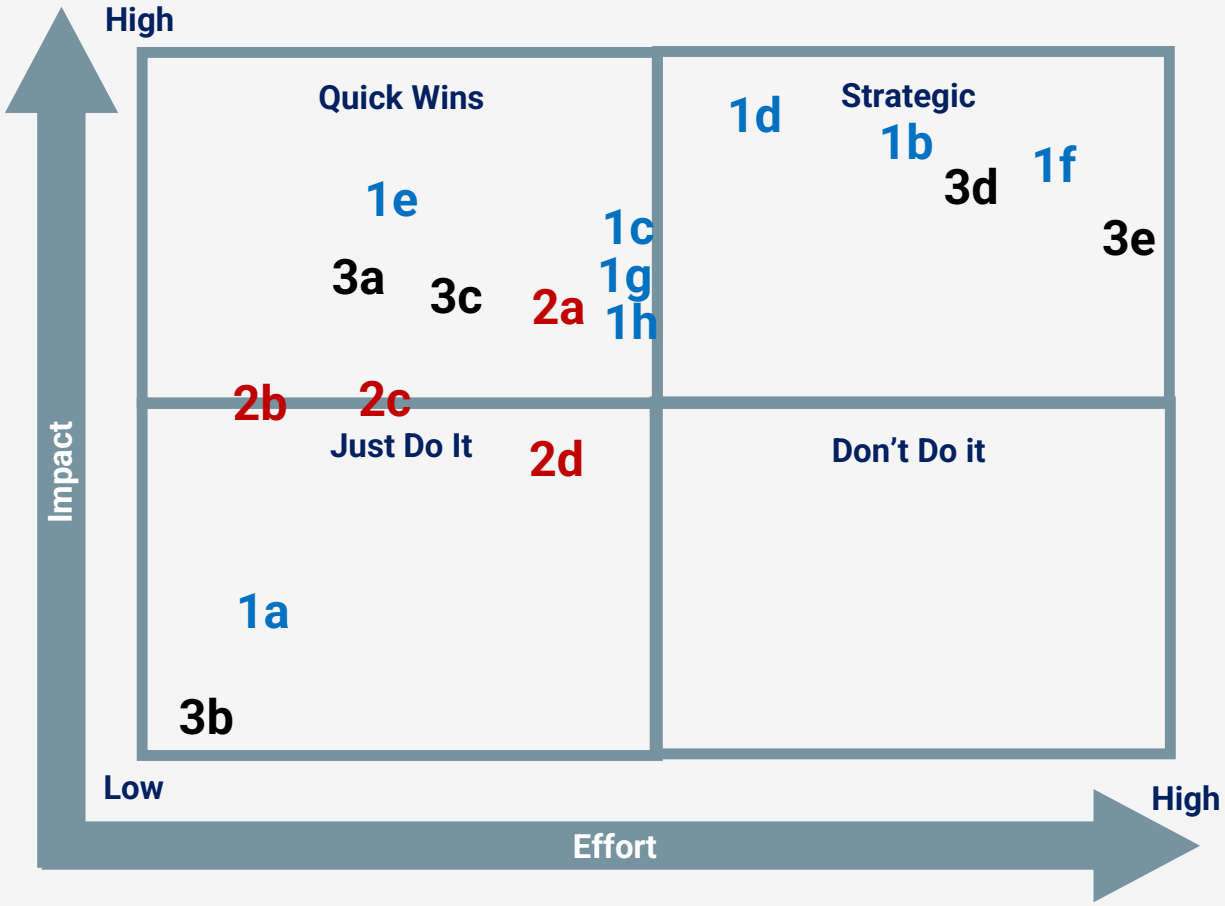
COMMENTS

Spools of wire is using a Supermarket approach with the operator driving the build based on empty slots.



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Project Prioritization Matrix



Priorities need to be considered in concert with each other.

Key Projects Identified

1	Problem Solving - PARETO Thinking
1a	Safety
1b	Kaizen Methodology
1c	TPM/SMED
1d	CAPEX Acquisition Criteria
1e	Tooling Diagnostics
1f	Machine Modification
1g	3P Design
1h	Unlinked Processes
2	SIOP - Demand Management
2a	Daily Management
2b	SQDC Focus & Hierarchy
2c	Leader Standard Work
2d	Employee Engagement
3	FAKE Lean
3a	Standard Work & Work Standards
3b	Standard Lexicon
3c	Respect for Standards
3d	Internal & Supplier Quality
3e	ABS Organization

Priority I Priority II Priority III

XYZ Facility - Financial Opportunities - Detail

With Rigourous Application of Lean Principles & Methodologies

USD $$$

USD (000's)

Item	Impact Area	Opportunity	Assumptions	Per Data Request	Improvement Impact Year 1	Year 2	Year 3	Total	3 Year Summary P&L Impact	Cash Flow Impact
Productivity	P&L + Cash Flow	Rigourous application of Standard Work with proper Cell Design	10 % Improvement per Year on Labor & Fringes assuming no wage increase and flat sales volume	$8,896	$890	$801	$721	$2,411	$18,651	$26,296
Quality	P&L + Cash Flow	Rigourous Quality Improvement program focusing on SUPPLIER, INTERNAL & EXTERNAL quality.	33% Minimum improvement annually in the Cost of (Poor) Quality. Assume COPQ = 10% of Cost of Production	$19,768	$6,524	$4,371	$2,928	$13,823		
SG&A	P&L + Cash Flow	n/a	Assume no SG&A Improvements		$0	$0	$0	$0		
Inventory Carrying Costs	P&L + Cash Flow	Reduction of carrying costs due to inventory reduction.	Inventory Carrying Costs @ 15% of Inventory Value	$2,972	$446	$825	$1,147	$2,417		
Inventory (Lead-time Reduction)	Balance Sheet (Cash Flow)	Right-size inventory with application Lean and Supply Chain Optimization	15% Improvement in inventory investment per year assuming flat sales volume (Represents ~12% improvement per year in Inventory Turns)	$19,813	$2,972	$2,526	$2,147	$7,645		
			Inventory Turns:	9.98	11.27	12.84	14.71			

Summary		Year 1	Year 2	Year 3	Total
EBIT Impact	Before Tax	$7,859	$5,996	$4,796	$18,651
Cash Flow Impact	P&L + Balance Sheet	$10,831	$8,522	$6,943	$26,296
Estimated Flat Net Revenue Based on Data Request	$273,800	$273,800	$273,800	$273,800	
Operating Profit / (EBIT) based on Data Request	$59,688	$ 67,547	$73,543	$78,339	
Operating Profit % (EBIT)	21.8%	24.7%	26.9%	28.6%	

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